Object Tracking: Attention over Time – how vision systems learn to remember

3 November 2025

v. 2.1

Course Goals

Object Tracking & Data Fusion

From detection to continuous perception — exploring how systems learn to keep an eye on the world.

Prompt Engineering

A quick detour on designing with language between human and machine.

Follow the white rabbit

Compare different Tracking algorithms, which tracker do you trust the most?

Why Deep Learning loves GPU’s

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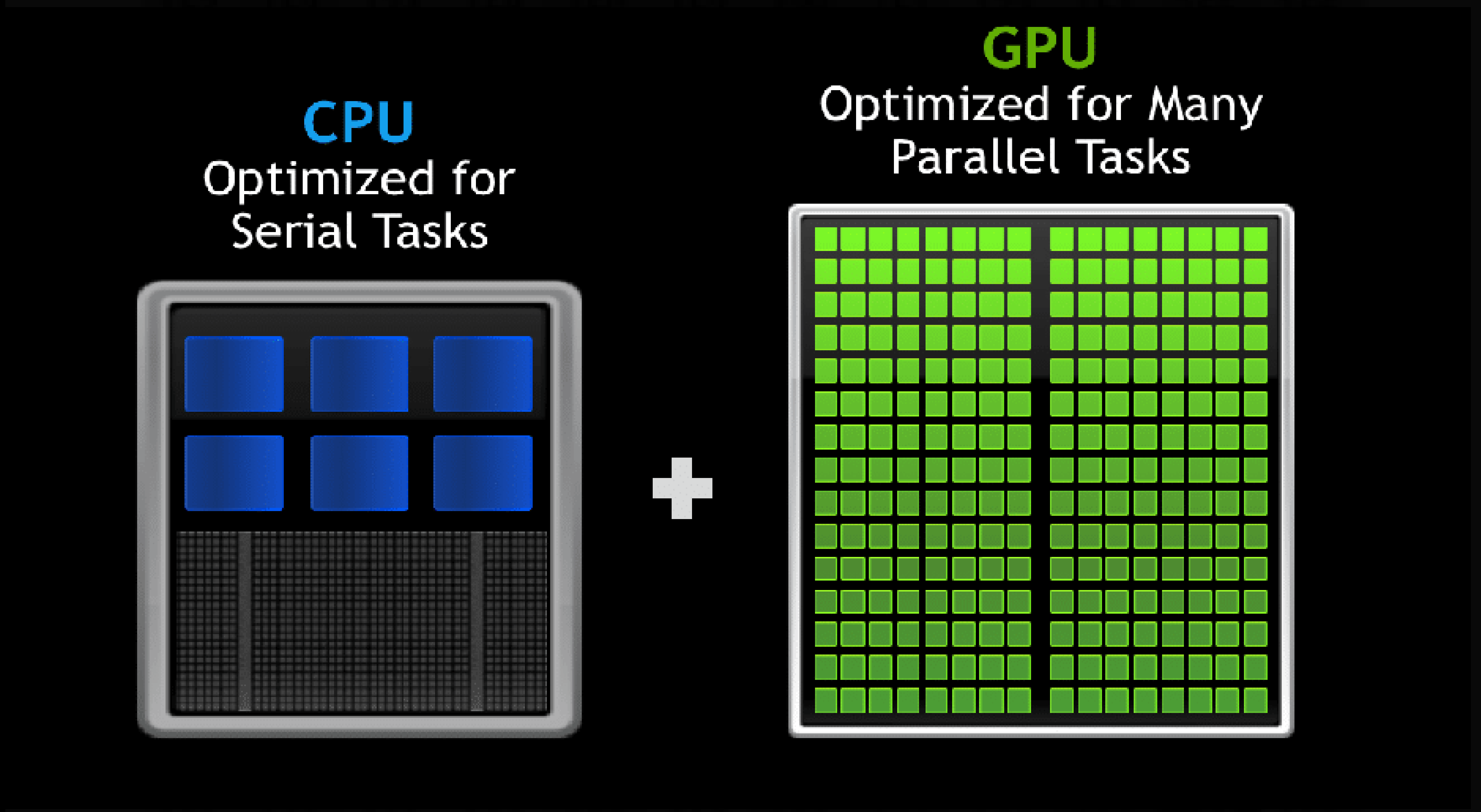
WhyDeepLearningloves GPU’s

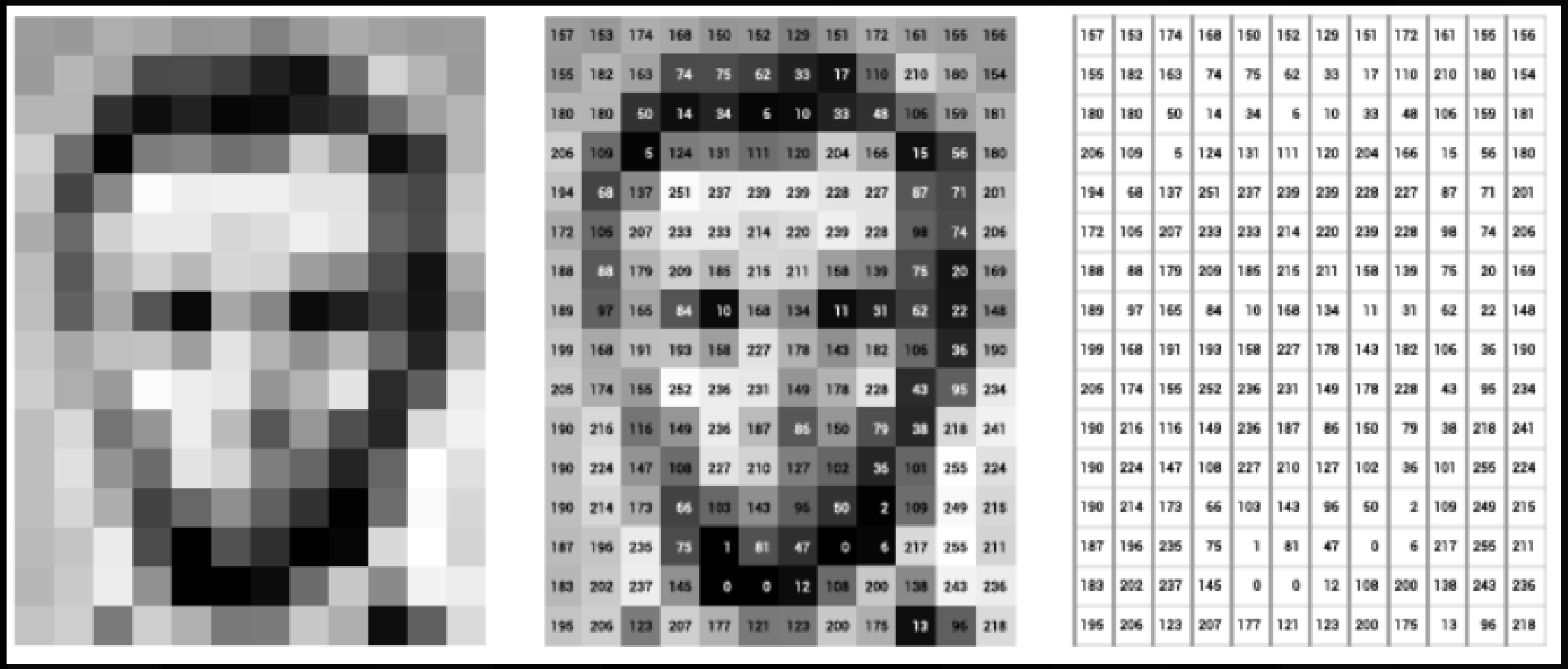
1.DeepLearning=MillionsofParallelCalculations
Everyneuronconnectioninvolvesmultiplicationsand additions—andCNNsapply*thousands of filters*over *millions of pixels.*

2.GPUsAreBuiltforParallelism

- CPUs=fewpowerfulcores(goodatlogic,control)
- GPUs=thousandsofsmallercores(goodat repeatingthesamemathfast)
- Idealformatrixopslikeconvolutionsand backpropagation

3.TheResult:
🧠Deeplearningtrains*faster*,scales*bigger*,and *converges better*whenrunningonGPU.





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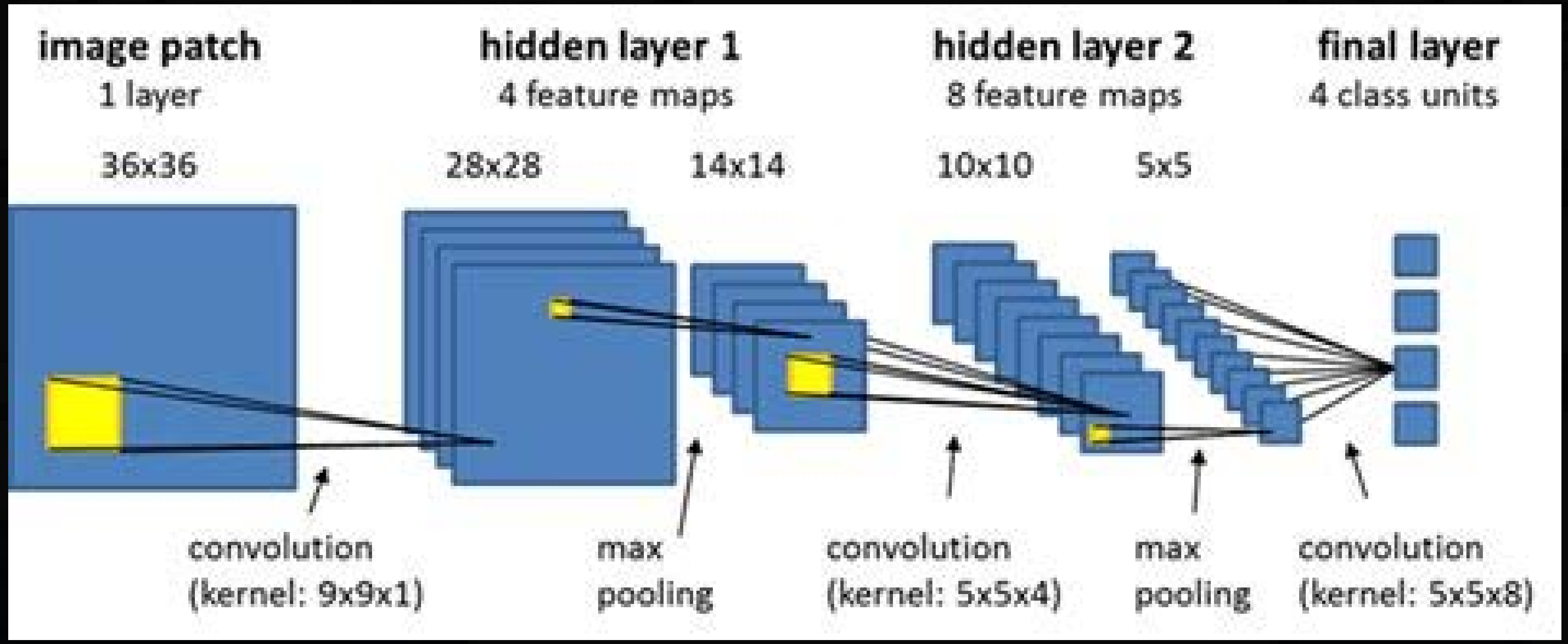

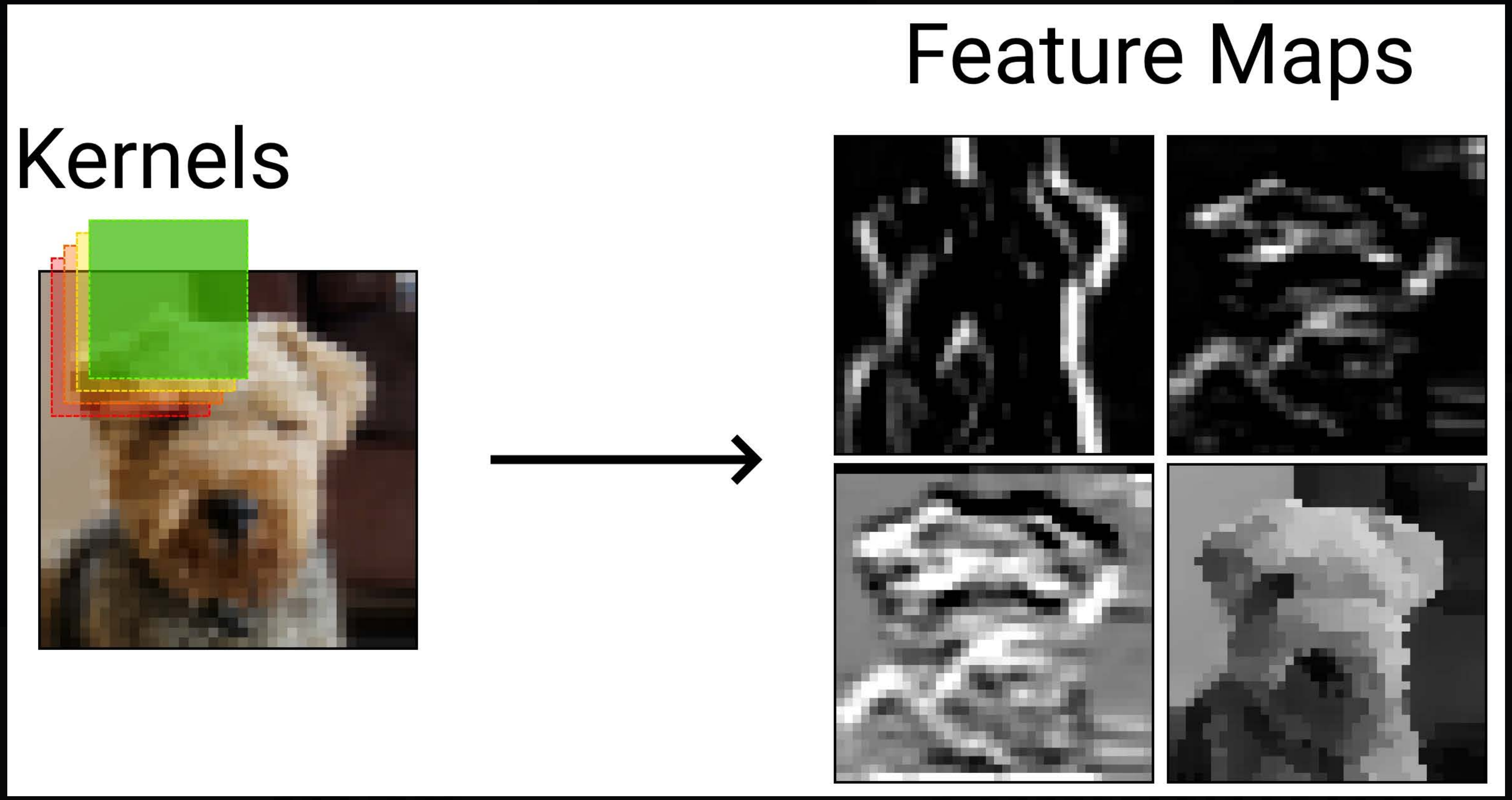

Tracking

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Tracking: Monitoring the movements and changes of an object over time

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Object Detection

identifiesandlabelsobjectsineach individualframe=Answers“what”

Object Tracking

keepsthesameobject’sidentityacross frames.Reliesonspatialcontinuity **(position,size,motion)**andsometimes appearancefeatures=Answers“where andwhen?”

Pipeline:
Detect->FocusonROI(regionofinterest) ->Rememberovertime

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Sensors → Machine Learning → Real-Time Data Processing

Sensors

Time-based sensors, such as RADAR, LIDAR or Cameras, provide the data to track objects.

Machine Learning

ML algorithms analyze patterns in the movement and predict future locations

Real-Time Data Processing

The ability to process data in real-time is crucial for effective tracking

ClassicalTrackersinOpenCV(LEGACY/OBSOLETE)

Tracker	MainIdea	Speed/Accuracy	Notes
KCF	Useskernelizedcorrelationfiltersonfeatures	Fast/Medium	Struggleswithocclusion
CSRT	Optimizedcorrelationfilterwithcolor&scaleadaptation	Medium/High	Robustbutslower
MOSSE	Adaptivecorrelationingrayscale	VeryFast/Low	Greatforreal-timedemos

Allareinitializedwitharegionofinterest(ROI)andupdatedeachframebasedonappearancecorrelation.

Deeptrackersintegratedetection+re-identification+motion modeling.

- **DeepSORT**=SORT+CNNembeddingsforappearance.
- **ByteTrack**=Useslow-confidencedetectionstoreduceID switches.
- **SiamMask/SiamFC**=Siamesenetworkslearnsimilarity betweentemplateandcurrentpatch.

Pipelineexample:

Video →Detector(YOLO)→ FeatureEncoder→ KalmanFilter→ IDAssignment

The math heart of tracking — predict → measure → correct

Kalman Filters

Predicts the object's next position based on motion equations, then corrects the estimate using new measurements.
Efficient for linear motion and Gaussian noise.

Particle Filters

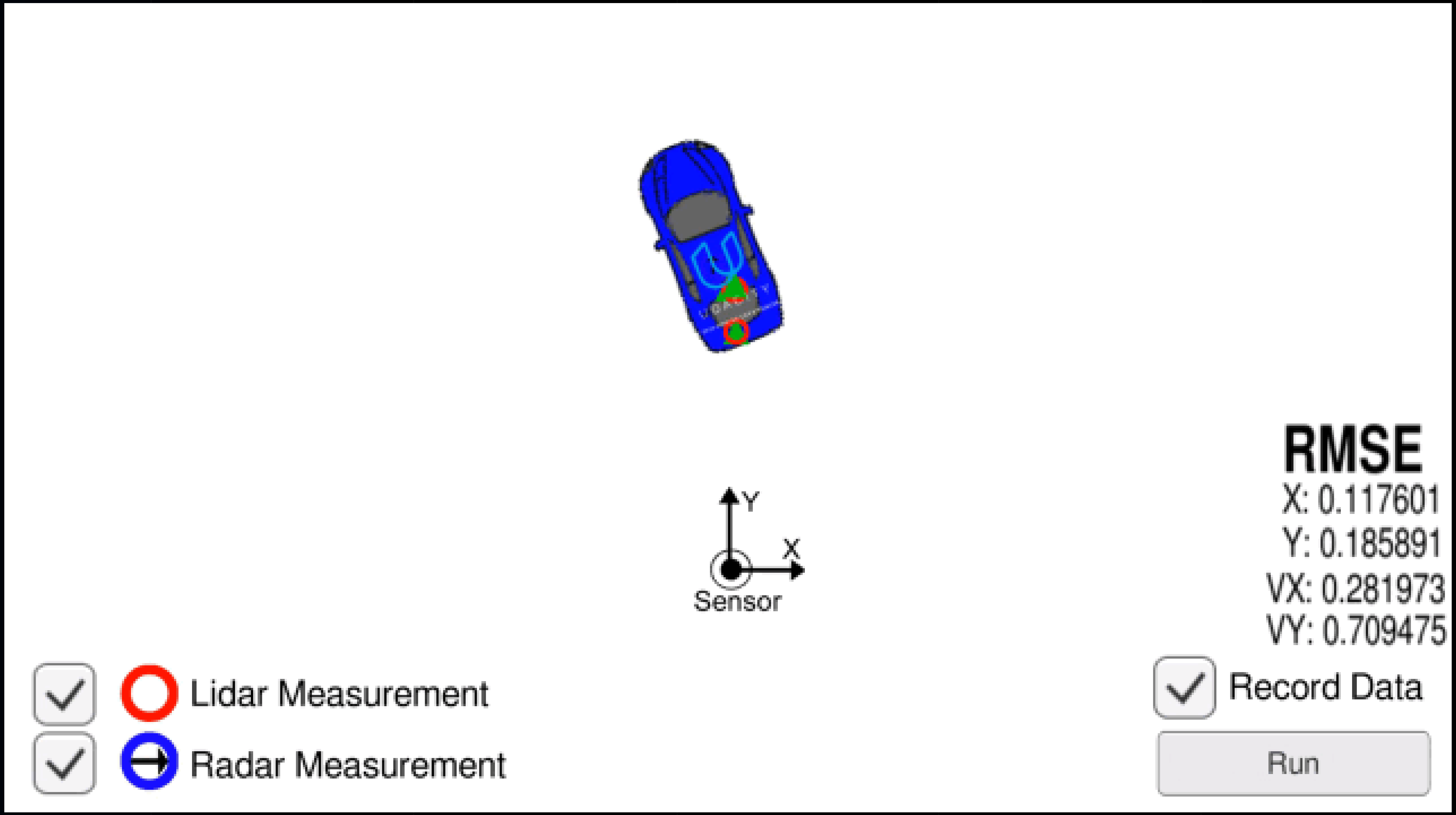
Uses many weighted "particles" to represent possible object states.
More flexible for nonlinear or uncertain motion.

Optical Flow

Estimates pixel-wise motion between consecutive frames based on changes in brightness patterns.
Often used for background motion or gesture tracking.

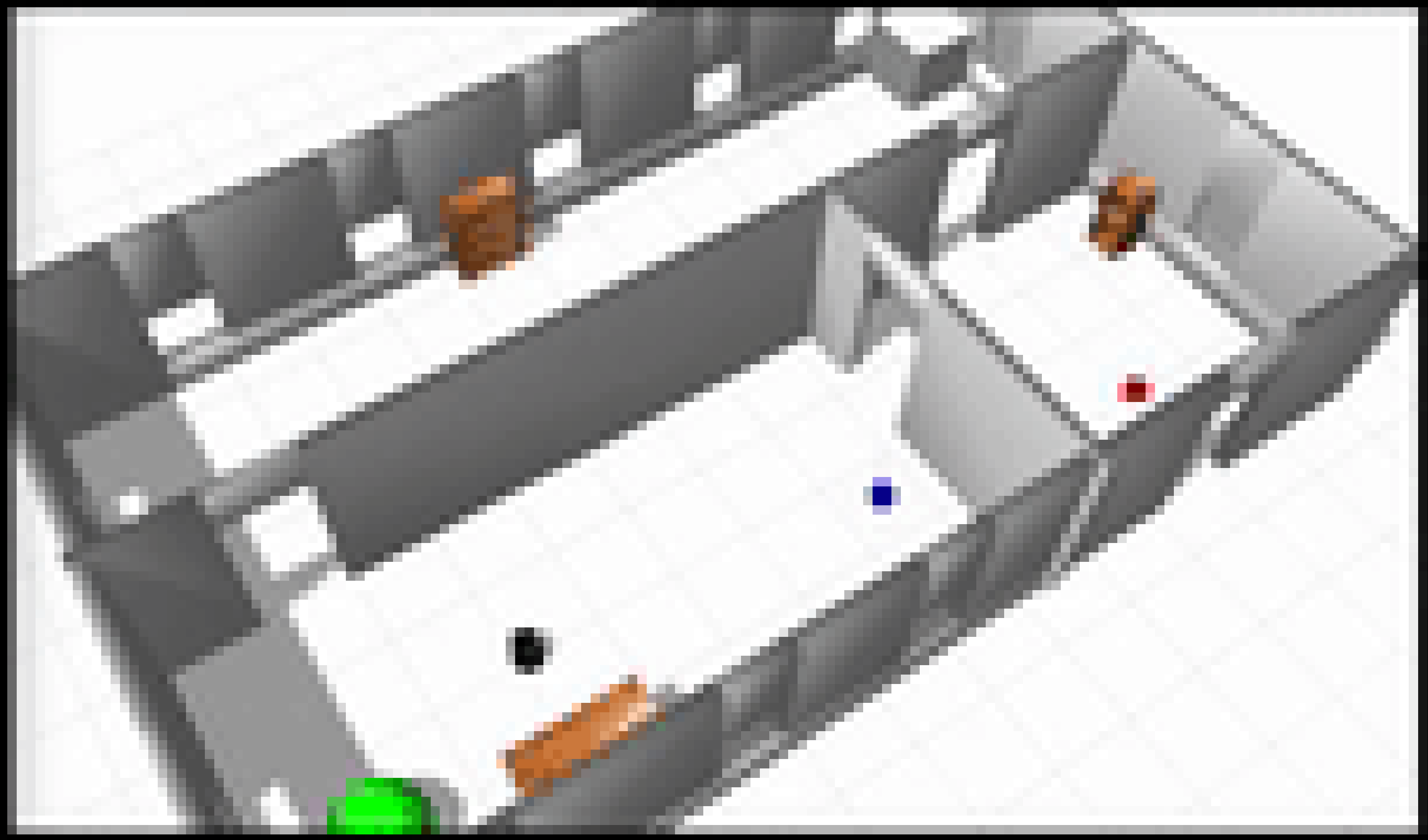
Multiple Object Tracking

Assigns consistent IDs to several detections over time using prediction and data association.

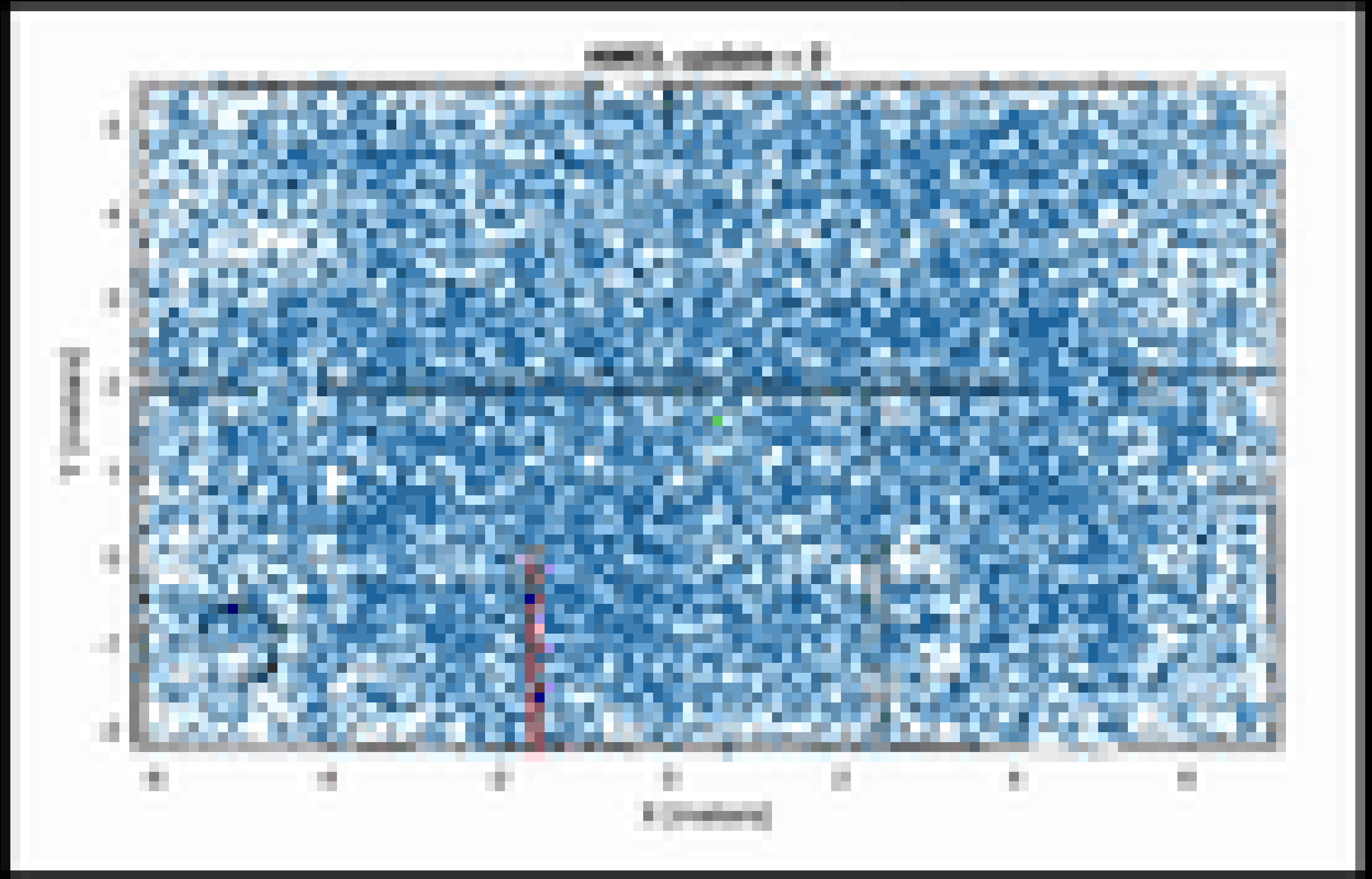


Y
X
Sensor
RMSE
X: 0.117601
Y: 0.185891
VX: 0.281973
VY: 0.709475
Lidar Measurement
Radar Measurement
Record Data
Run

Simulated real environment



Estimated state from particle filter



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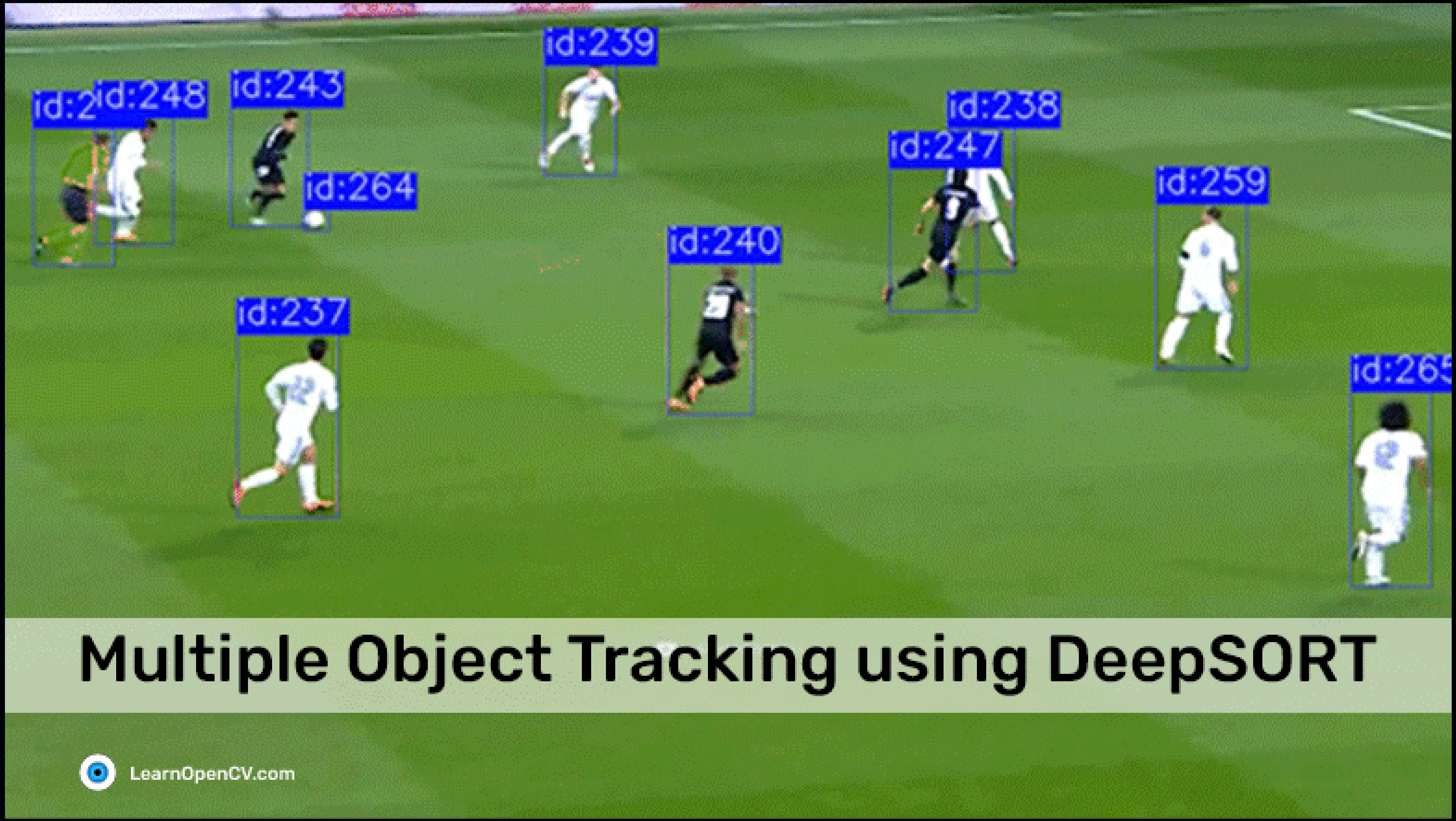

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Tracking Challenges

- **Occlusion:**objectdisappearsbehind another.
- **Scalechange:**objectmovescloser/ farther.
- **Lightingvariation&motionblur**:pixel patternsshift.
- **IDswitch:**trackerconfusesidentities.

MOTChallengedatasetsandevaluation metrics(MOTA,IDF1):https:// motchallenge.net/

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Person Localisation & **Tracking**

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Person Localisation & Tracking

Personlocalizationandtrackingreferstotheprocess ofdetectingthepresenceandlocationofapersonin animageorvideoframeandsubsequentlytracking theirmovementsacrossmultipleframes.

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Techniques & Applications behind Person Localisation & Tracking

Techniques

- ObjectDetection:Techniqueslike Haarcascades,HOG+Linear SVM,YOLO,FasterR-CNN,etc.
- Tracking:OpticalFlow,Kalman Filter,MeanShift,CAMShift,etc.
- DeepLearning:Convolutional NeuralNetworks(CNNs), RecurrentNeuralNetworks (RNNs),LongShortTermMemory Networks(LSTMs)

Applications

- **Surveillance**:Trackingpersonsof interestinsurveillancevideofootage.
- **SportsAnalysis**:Analyzing movementsandactionsofplayersin asportsgame.
- **Human-ComputerInteraction**: Detectingandtrackinguser movementsforinteractivesystems.
- **Robotics:**Helpingrobotsunderstand andnavigatetheirenvironment.

https://docs.opencv.org/3.4/d7/d53/tutorial_py_pose.html

Articulated **Body Tracking**

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Articulated Body Tracking

- Articulatedbodytrackingreferstotheprocess ofnotonlytrackingtheoverallmovementofa personbutalsothemovementandpositionof eachpartoftheirbody-liketheirhead,arms, torso,andlegs.
- Articulatedbodytrackingistypicallydoneusing poseestimationtechniques,whichcandetect thepositionandorientationofdifferentbody parts.Thisinvolvesdetectingkeypoints(or "joints")ontheperson'sbodyandthenusing thesekeypointstoconstructa“skeleton.”
- Theimageontheright-handsideshowsBody TrackingwithAppleARKit

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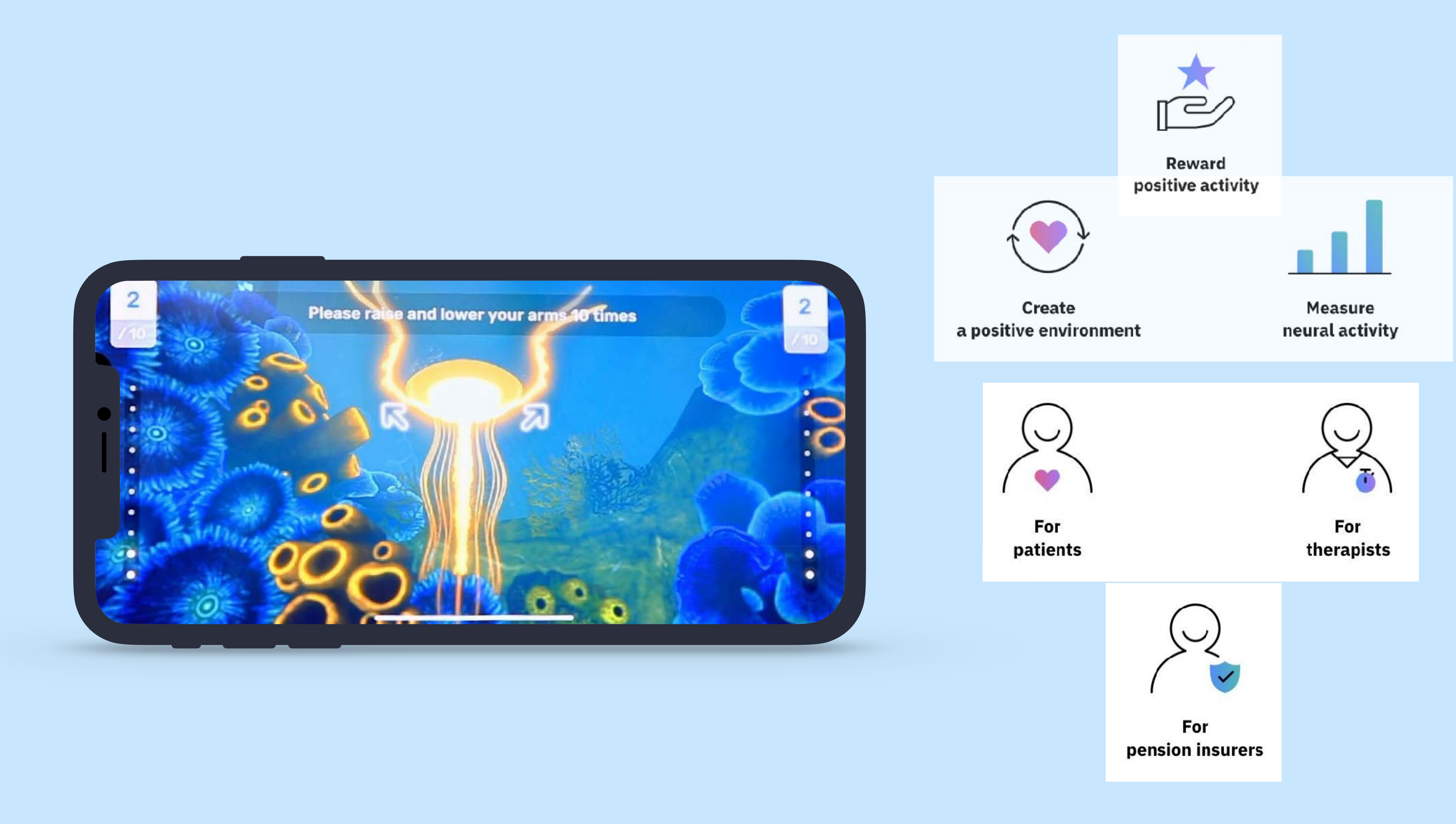

Techniques behind Articulated Body Tracking

- PoseEstimation:MethodslikeOpenPose, PoseNet,andDeepPose
- SkeletalTracking:Constructingaskeleton fromdetectedkeypoints,andtrackingthe movementofthisskeletonacrossframes
- DeepLearning:Techniqueslike ConvolutionalPoseMachines,Stacked HourglassNetworks,andPartAffinity Fields.

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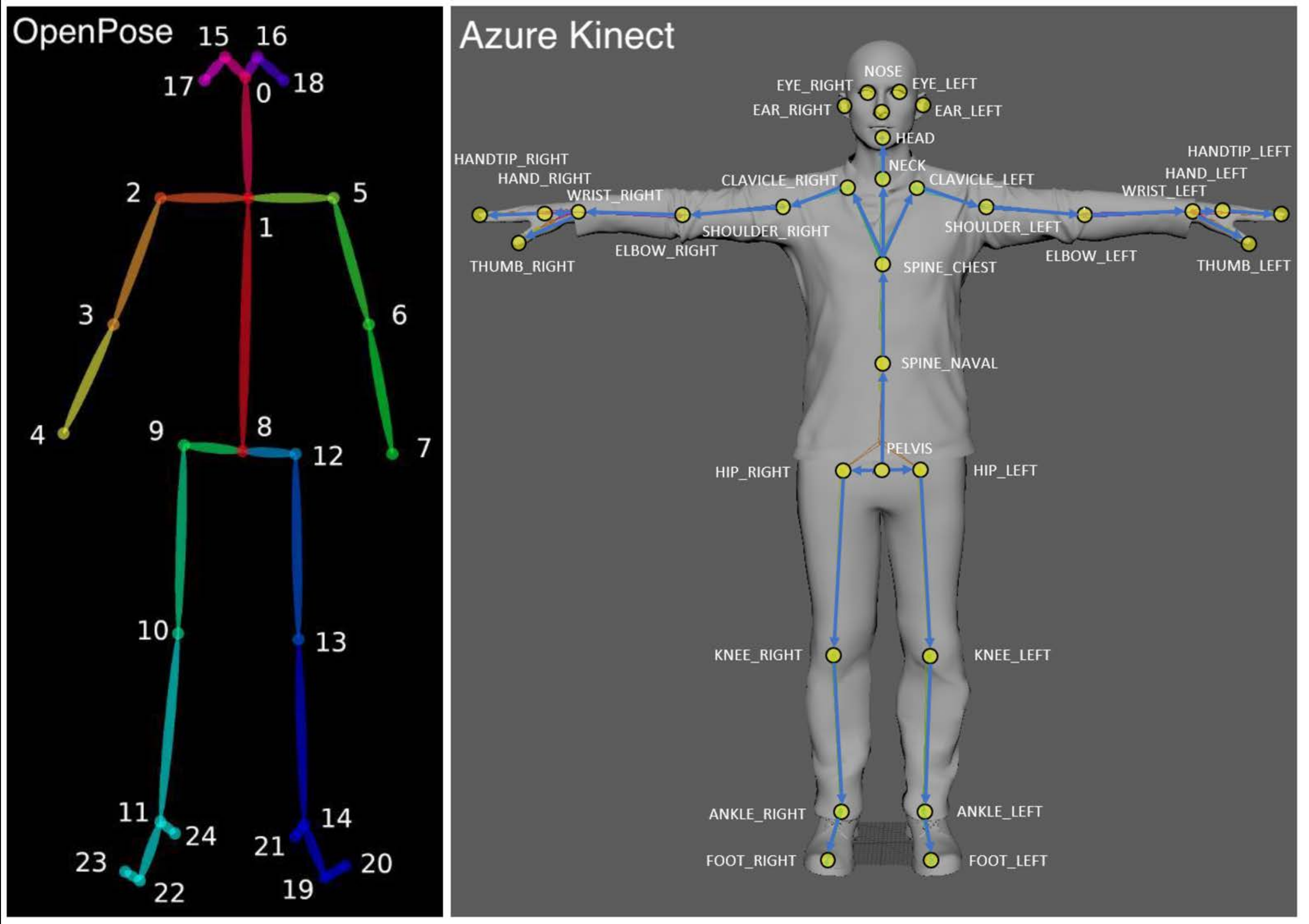
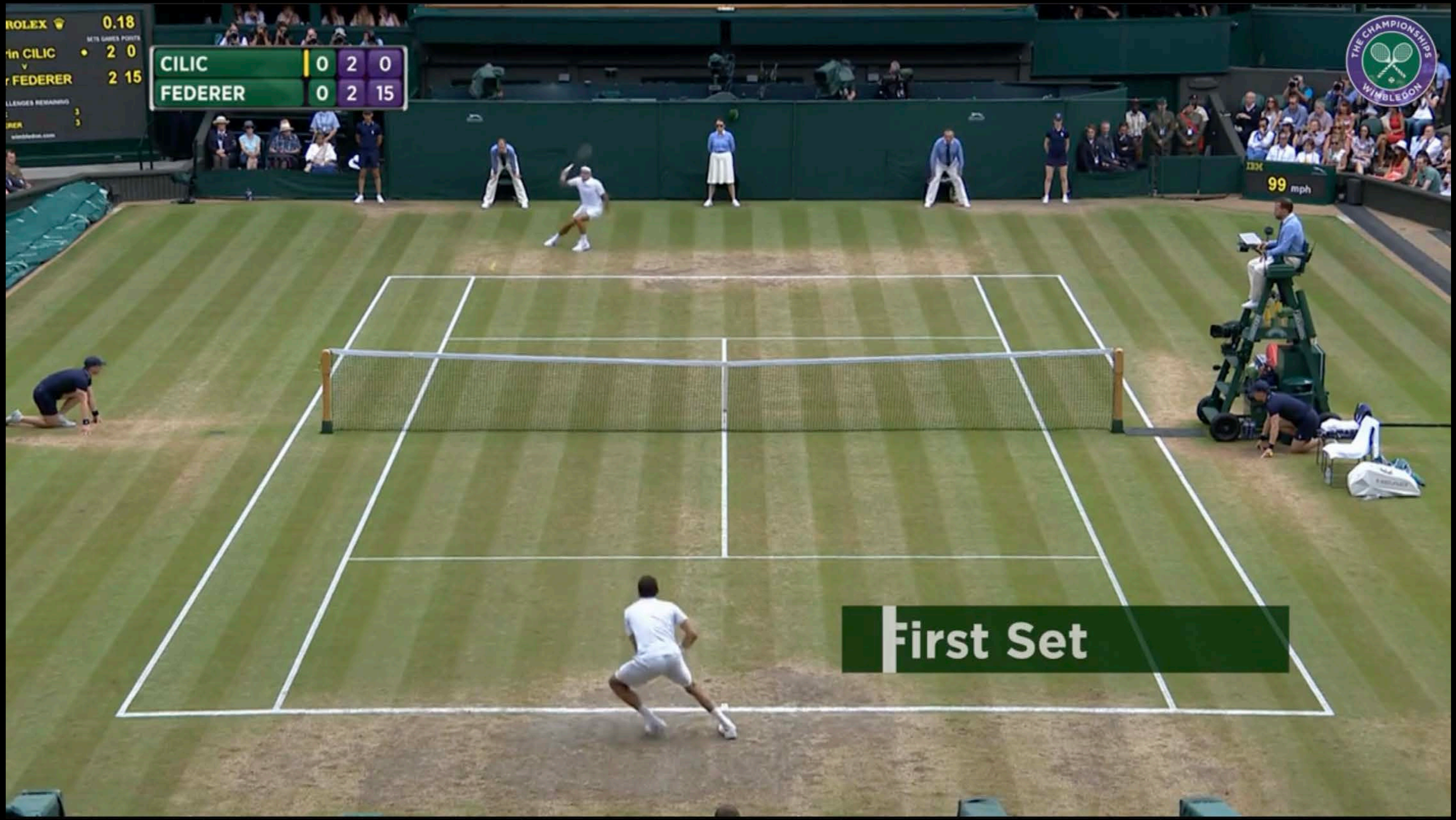
Applications

- MotionCapture:Capturinghuman movementsforanimationandgaming.
- SportsTraining:Analyzingbodypostureand movementsforsportstraining.
- HealthandRehabilitation:Trackingbody movementsforphysiotherapyand rehabilitation.
- Human-ComputerInteraction:Detecting andinterpretingbodygesturesfor interactivesystems.



Creating Match Highlights 2-minutes after a game ended



https://www.wimbledon.com/en_GB/video/media/0411310c2acc546edee3d24d2213822f.html

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Automated Highlights

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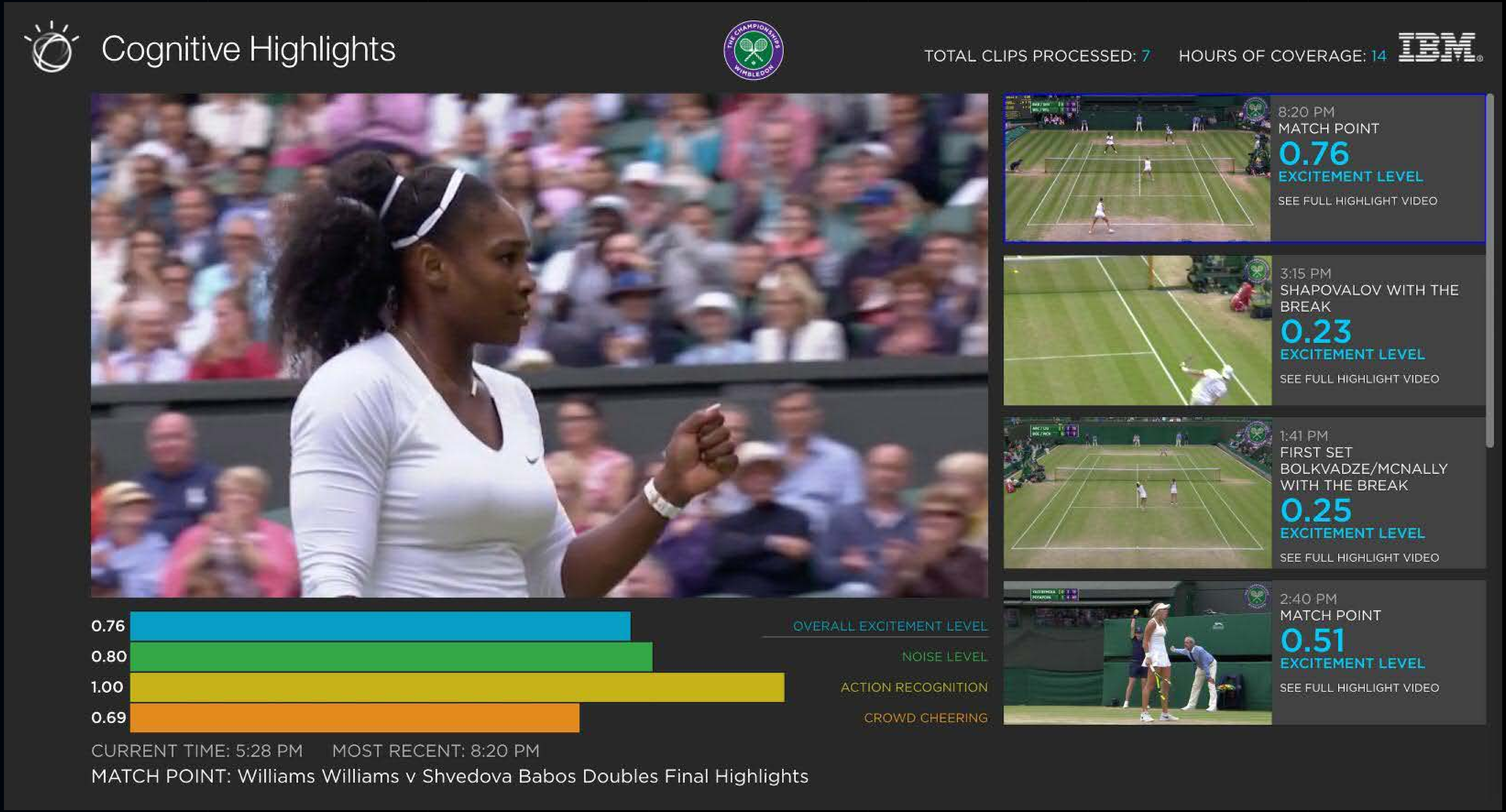

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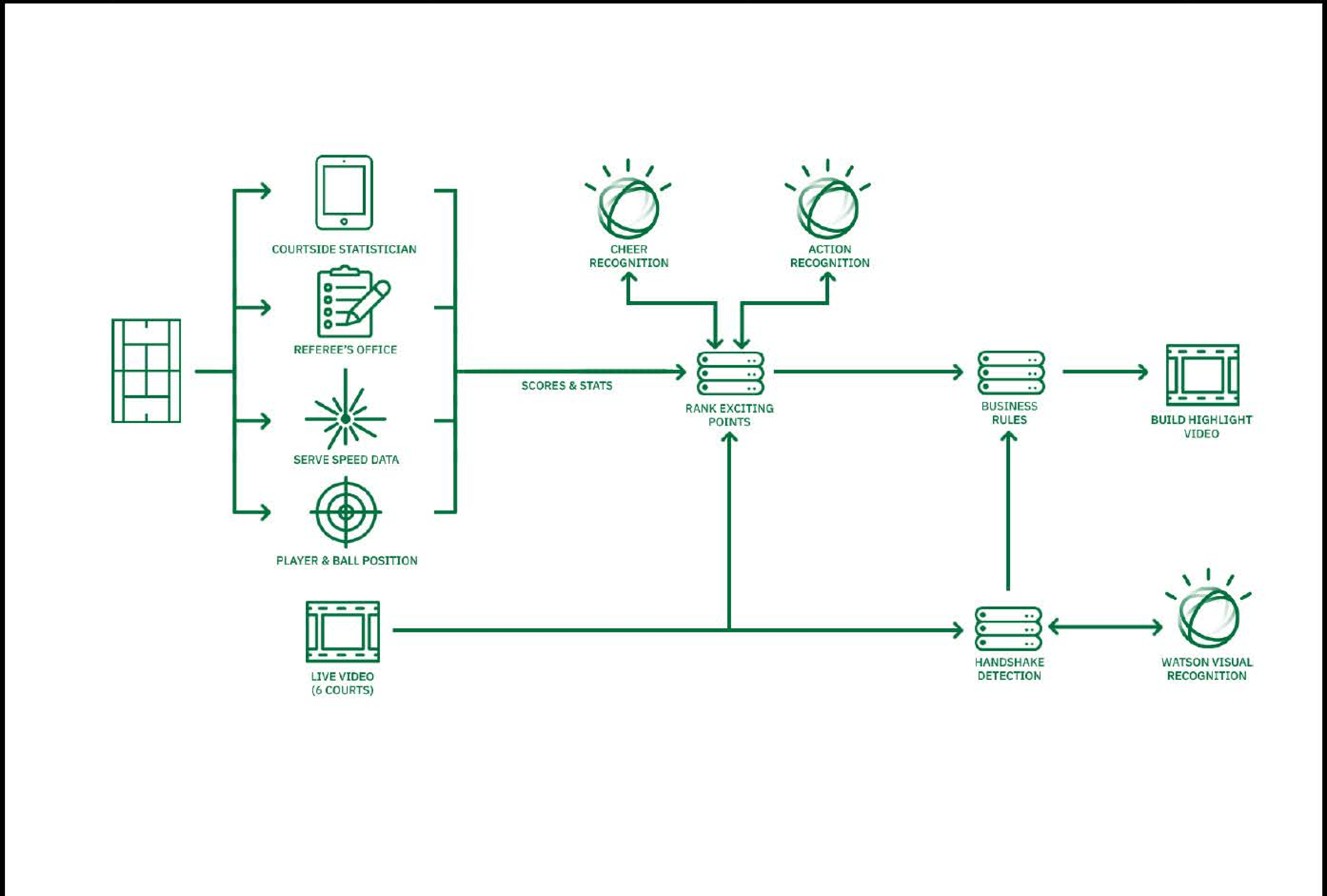
Putting different AI technologies together

1. Point Selection – The Watson AI system captured exciting moments using audio and video AI tools that analyze crowd cheering, action recognition (i.e., visuals of player behavior), and scoring data. The video segments are ranked and selected based on these different sources to produce the final highlight video for each match.

2. Clipping and fine-tuning – A Watson Visual Recognition classifier was built to ensure that only desired content was included in the final video. The system was trained to learn which broadcast shots to include and which to avoid (thus excluding any non-tennis action) and identify end-of-match handshakes to finish the highlight videos.

3. Production – Once points were selected, timestamps were determined and verified to ensure they were all in the correct order. Files were then handed off to a Highlights Asset Manager to add pre-agreed storytelling graphics, watermarks, and transitions.

4. Distribution – Content was uploaded automatically to the wimbledon.com content management system, making them quickly available for use. From there, the team could publish, share or download content as desired.



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Fox Sports FIFA Highlight Machine

TheHighlightMachineisa unique,fan-firstviewingand creatingplatform

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The Insight

1billionpeopletuneintoWorldCup.
Theaudiencewillbeyoungandglobal.

Andeveryfanhashisorherfavoriteplayer
orteam.

Yetbroadcasterstreateveryonethesame.

There'szeropersonalization.
It'sthesameoldone-wayconversation.

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It started with

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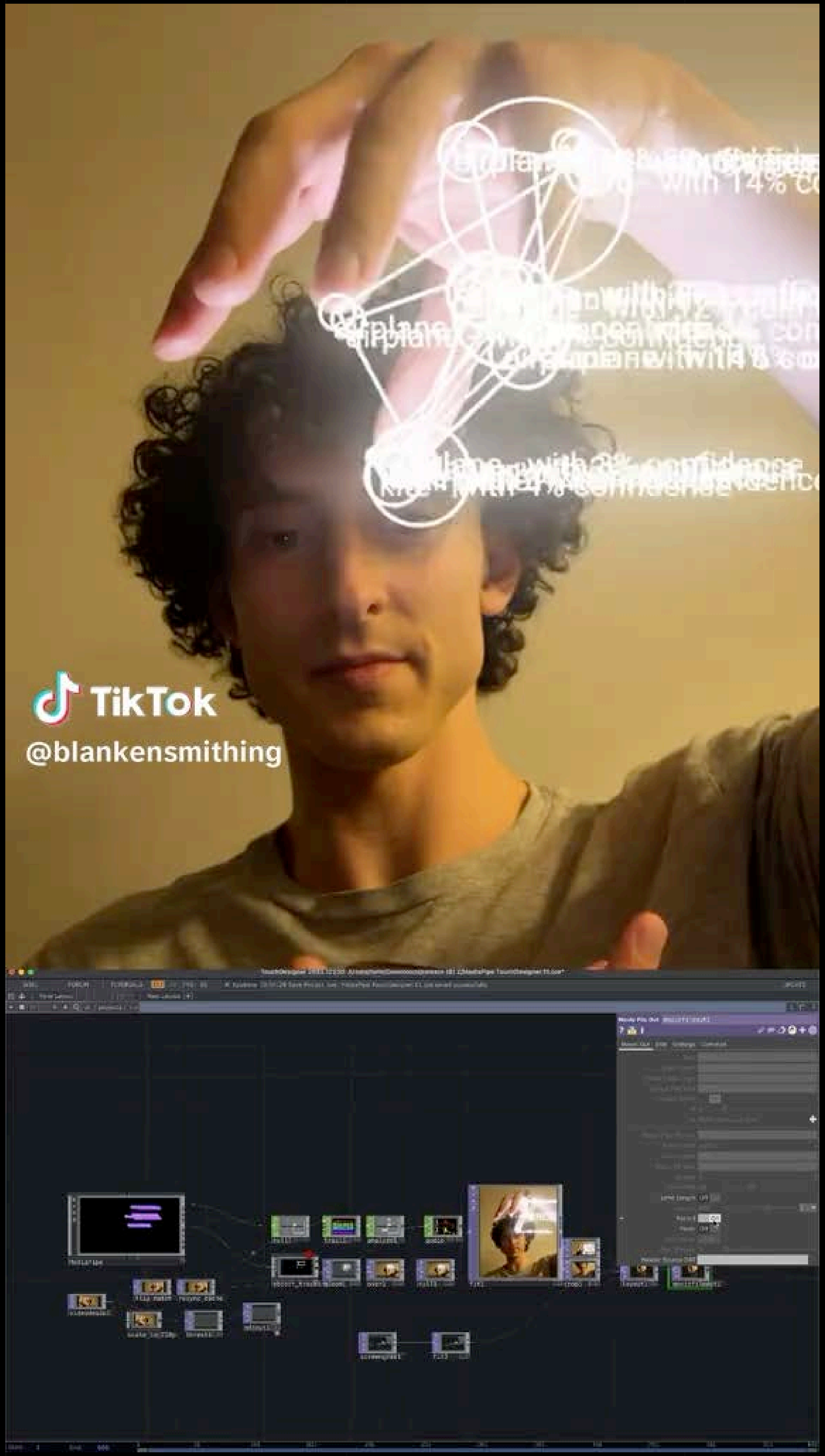

PracticalActivity

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Sensor Fusion: Enhancing Accuracy, Completeness, and Reliability in Modern Technology

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Data Fusion is the process of integrating multiple data sources to form helpful information that is more consistent and more accurate than the original data sources.

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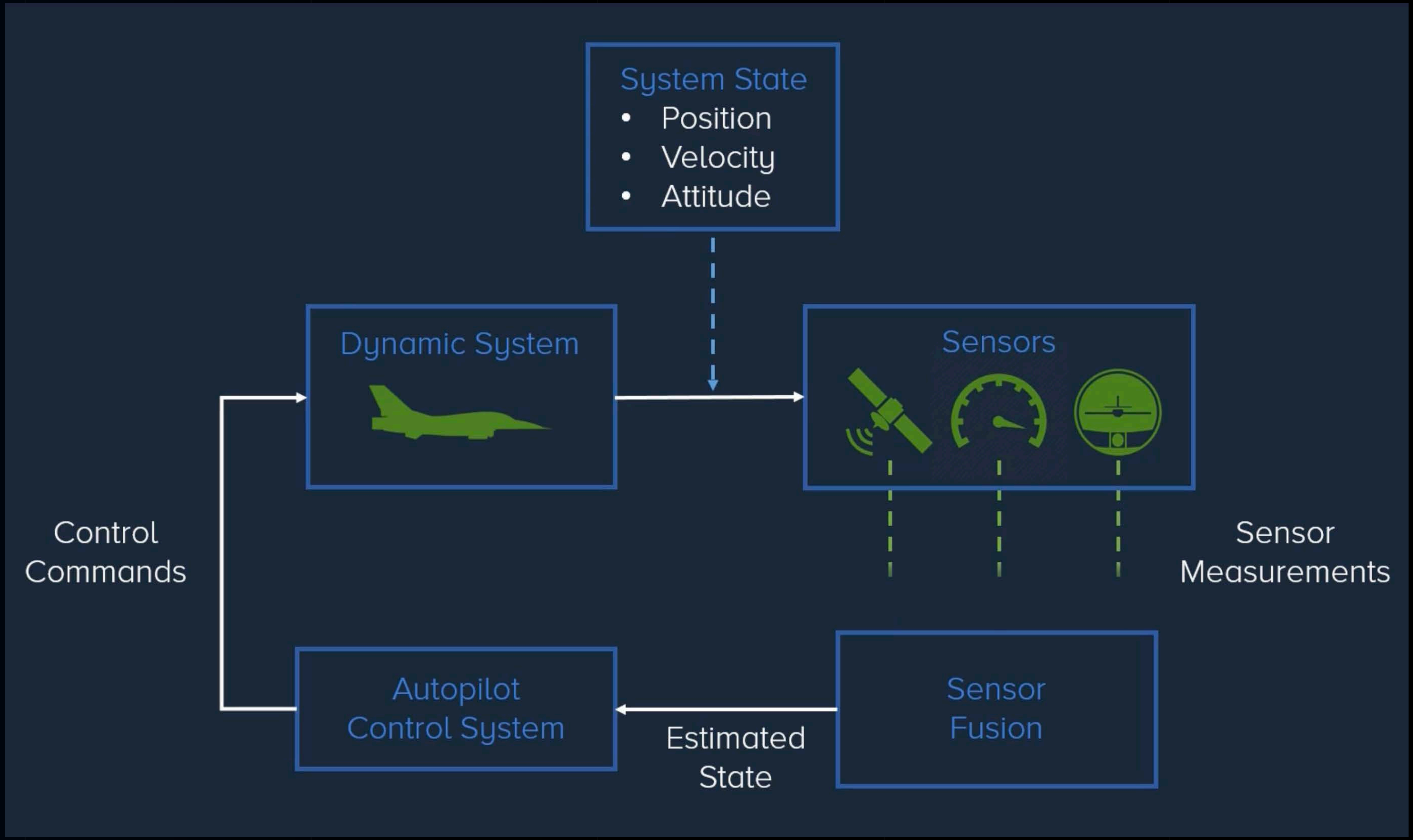
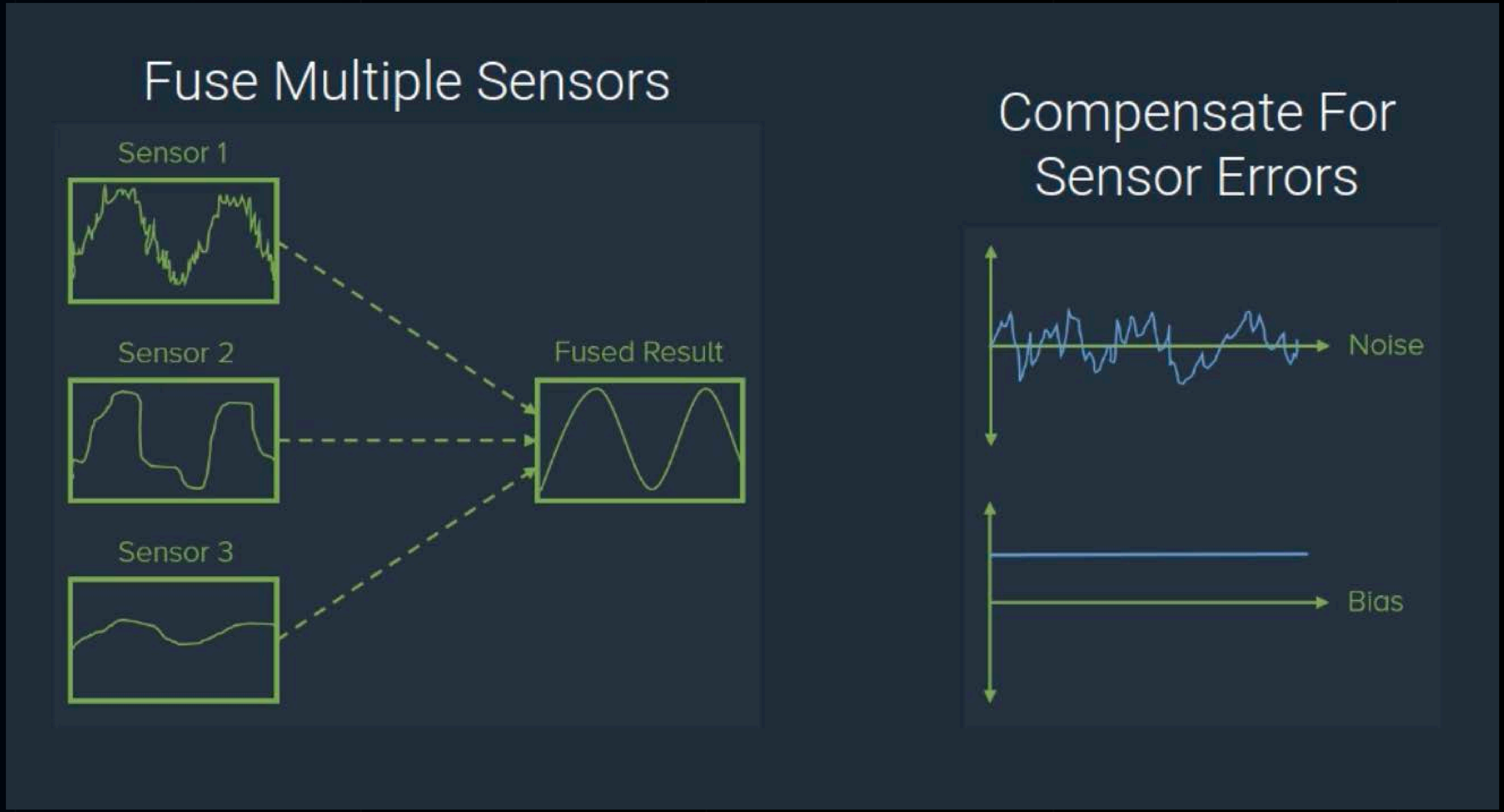
Combining sensor measurements in a way that the resulting fusion produces a more confident estimate than the individual measurements alone



- Dynamicsystem:Motionand dynamicsofanaircraft
- Stateofthesystem:Current properties(velocity,position, attitude)
- Autopilotcontrolrequires knowledgeoftheaircraft'sstate
- Sensorsattachedtotheaircraftto measurephysicalproperties (GPS,speed,altitude)

No sensor in real life is perfect, and any measurement it provides will be corrupted with errors such as noise and biases.

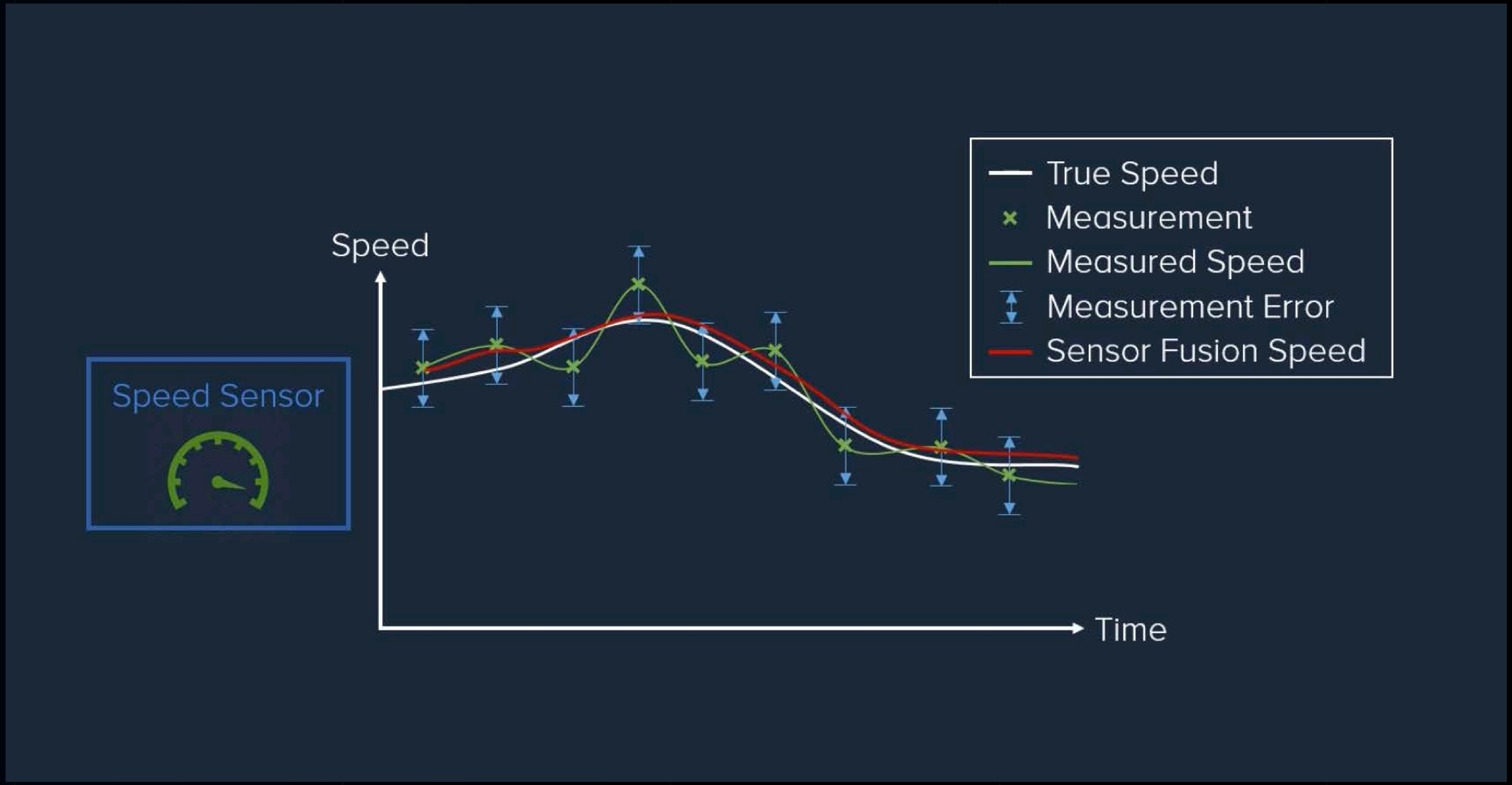


- LimitationsofIndividualSensors
- Nosinglesensorcanprovideall therequiredinformation
- Sensorfusionprocessneededto obtainacompletepictureofthe aircraft'sstate

BenefitsofSensorFusion

- ErrorCorrection:Real-lifesensors areimperfect,pronetonoiseand biases
- Fusionreducestheeffectsof errors
- Rawsensordataisuseless withoutfusiontocorrecterrors

Example: Speed Sensor Fusion and how fusion procides a better estimate of the true speed of an object



- ImpactofSensorFusionon AutomationandControl
- Increasedaccuracy,robustness, andreliability
- Automationsystemsrelyon accurateinformation
- Withoutfusion,automationcan bedangerous(e.g.,cruisecontrol inacar)

Understanding Data Fusion: Bayesian or Probabilistic Approach

- Datafusionisawidesubjectarea,focusingonone powerfulmethod:Bayesianorprobabilisticdata fusion.
- Encapsulatingthestateofadynamicsystemasa probabilitydistribution.
- Updatethedistributionwithsensormeasurements toobtaintheoptimalstateestimate.
- Probabilityframeworkcapturesuncertaintyand enhancesestimationaccuracy.

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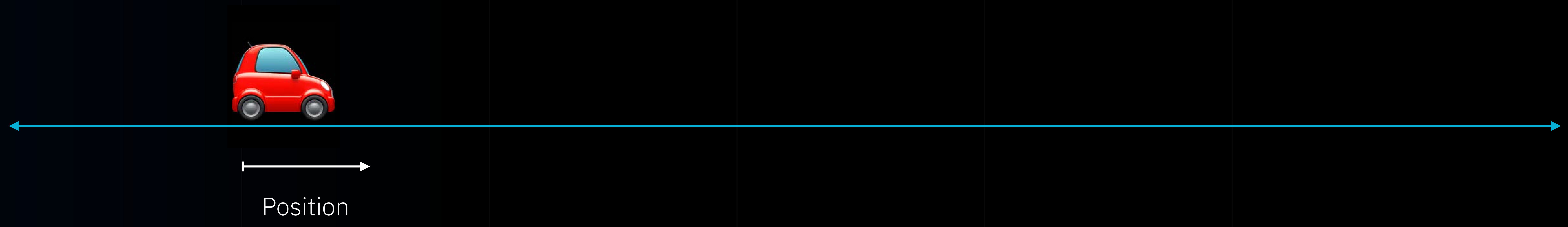
How Data Fusion Works

Carcantravelinonedirection,asseenhere,soitspositioncanshiftleftandright.
Ourgoalistoestimateortrackthecar'spositionasitmovesandevolvesovertime.



How Data Fusion Works

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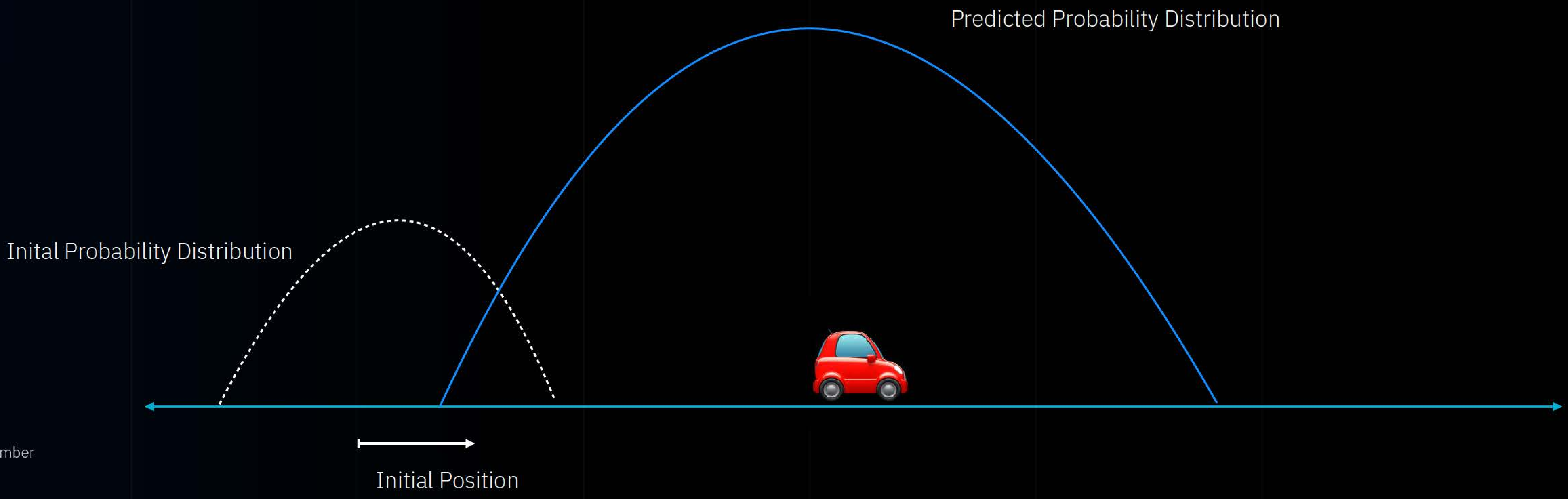
How Data Fusion Works



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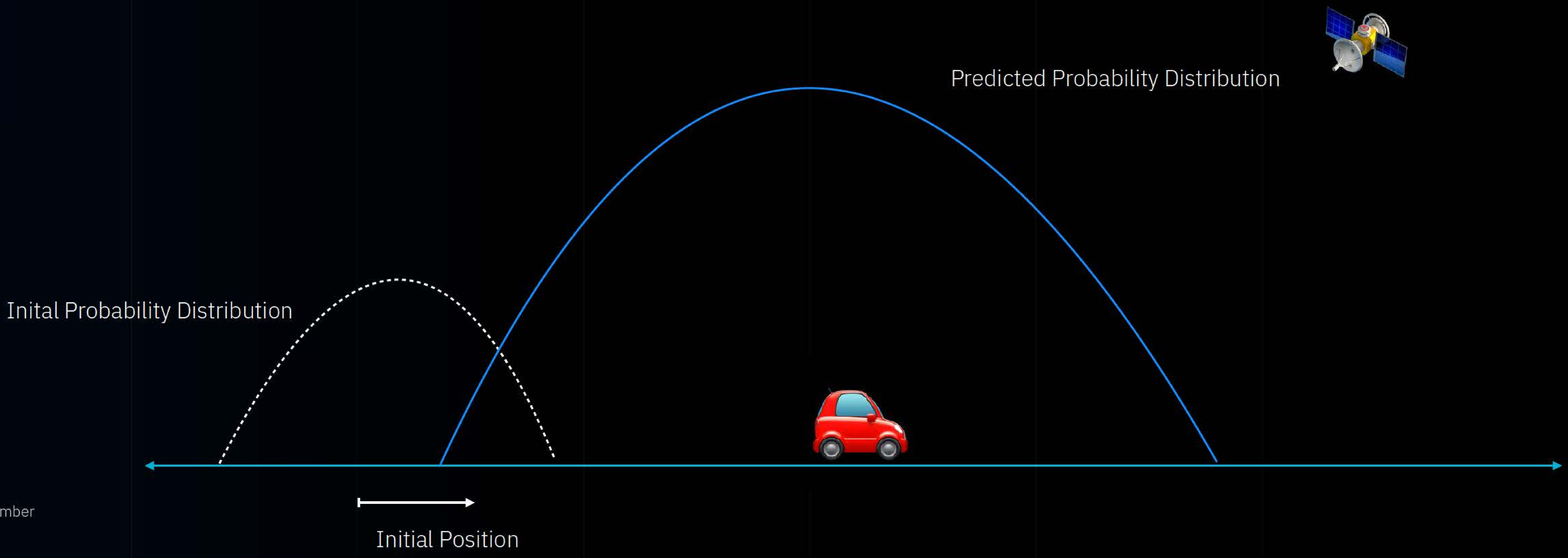
How Data Fusion Works



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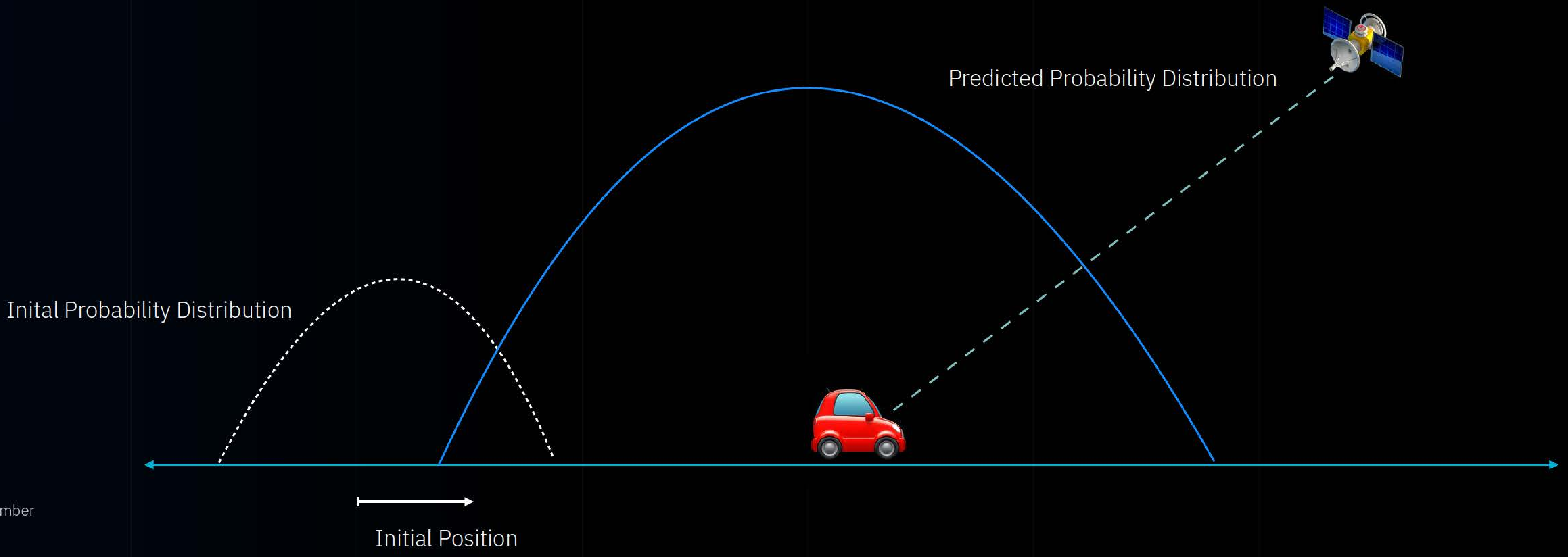
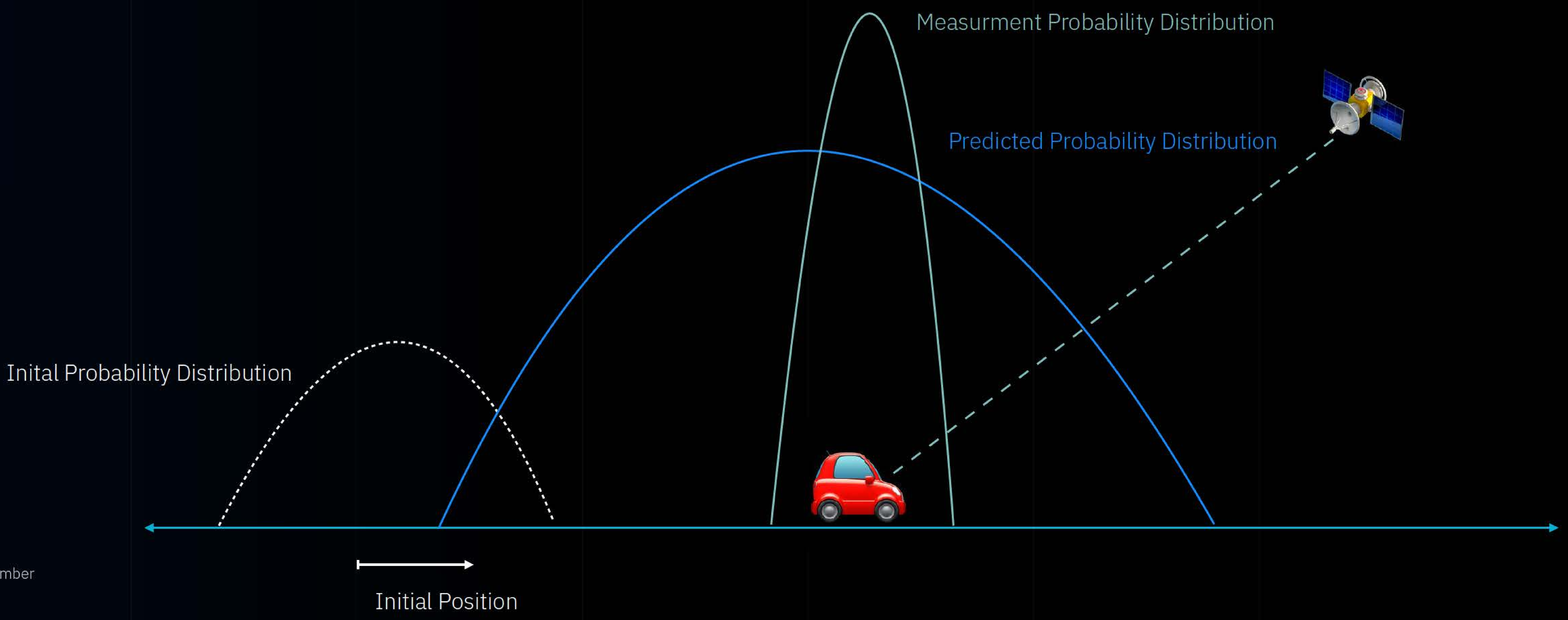
How Data Fusion Works



How Data Fusion Works



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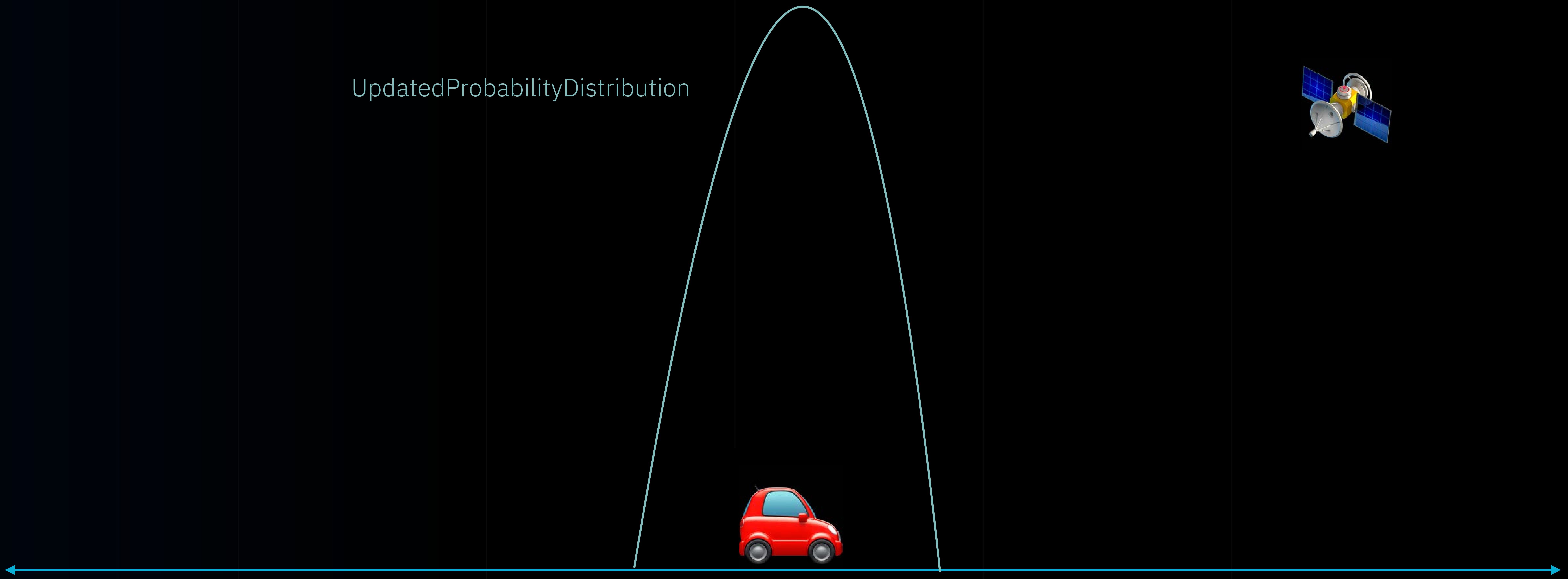
How Data Fusion Works



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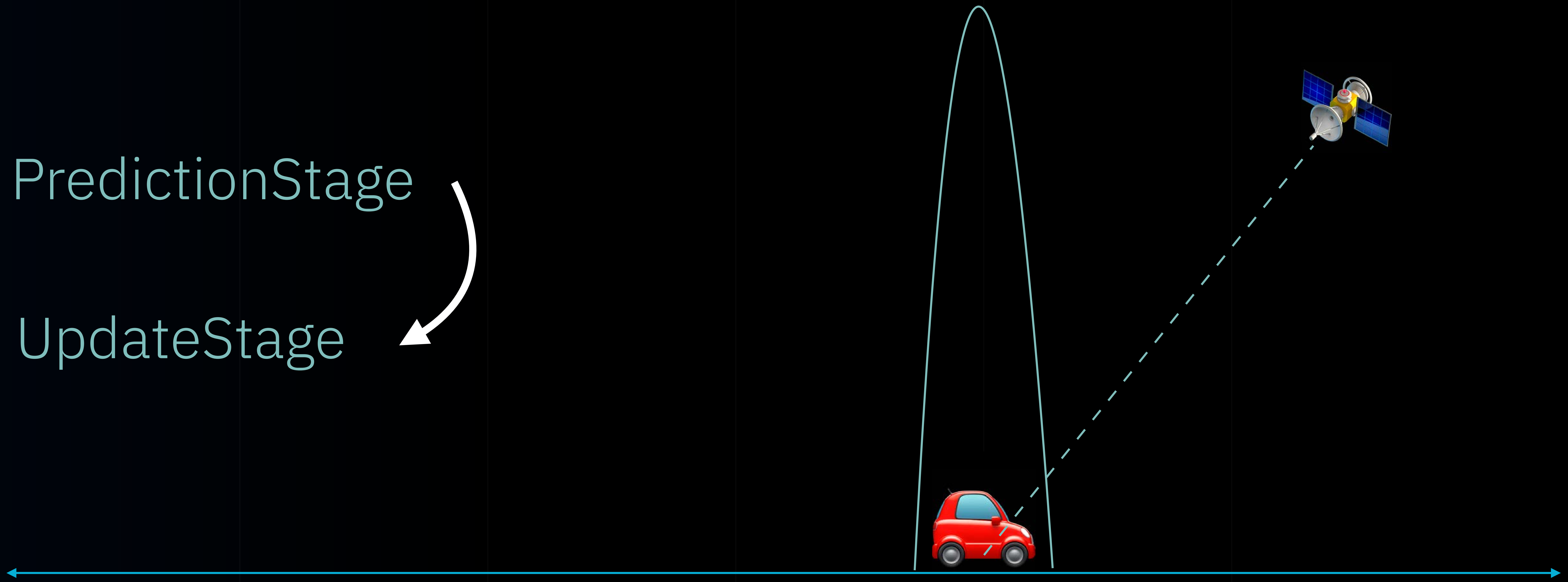
How Data Fusion Works

PredictionStage

UpdateStage



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Data Fusion Stages

Steps forward in Time
(Uncertainty Grows)

Prediction Stage

Update Stage

Updates current time
(Uncertainty
Shrinks)

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current estimation

measured value

$$\hat{X}_k = K_k \times Z_k + (1 - K_k) \cdot \hat{X}_{k-1}$$

Kalman Gain

previous estimation

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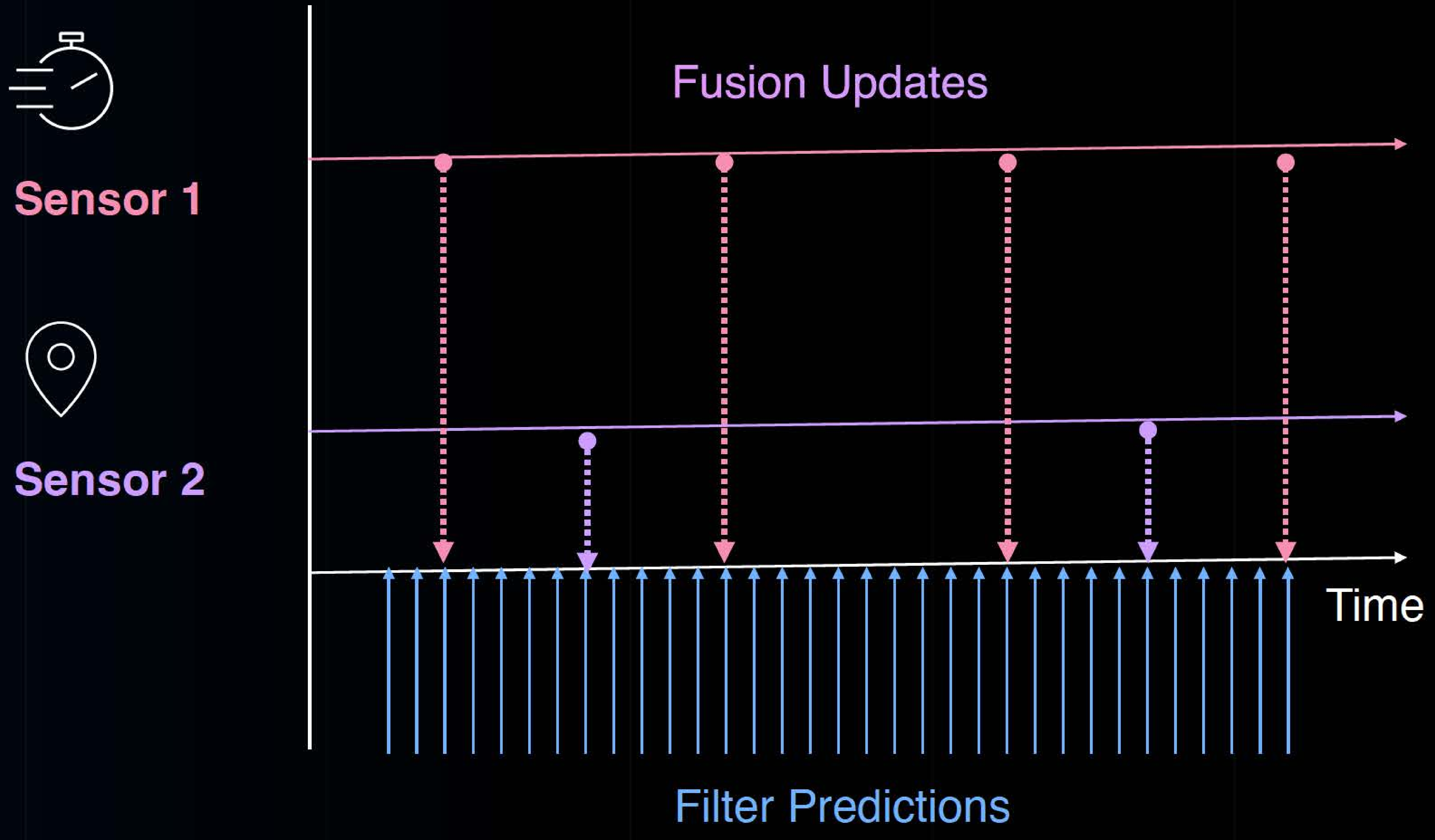
Fusion updates can be executed Non-Sequential



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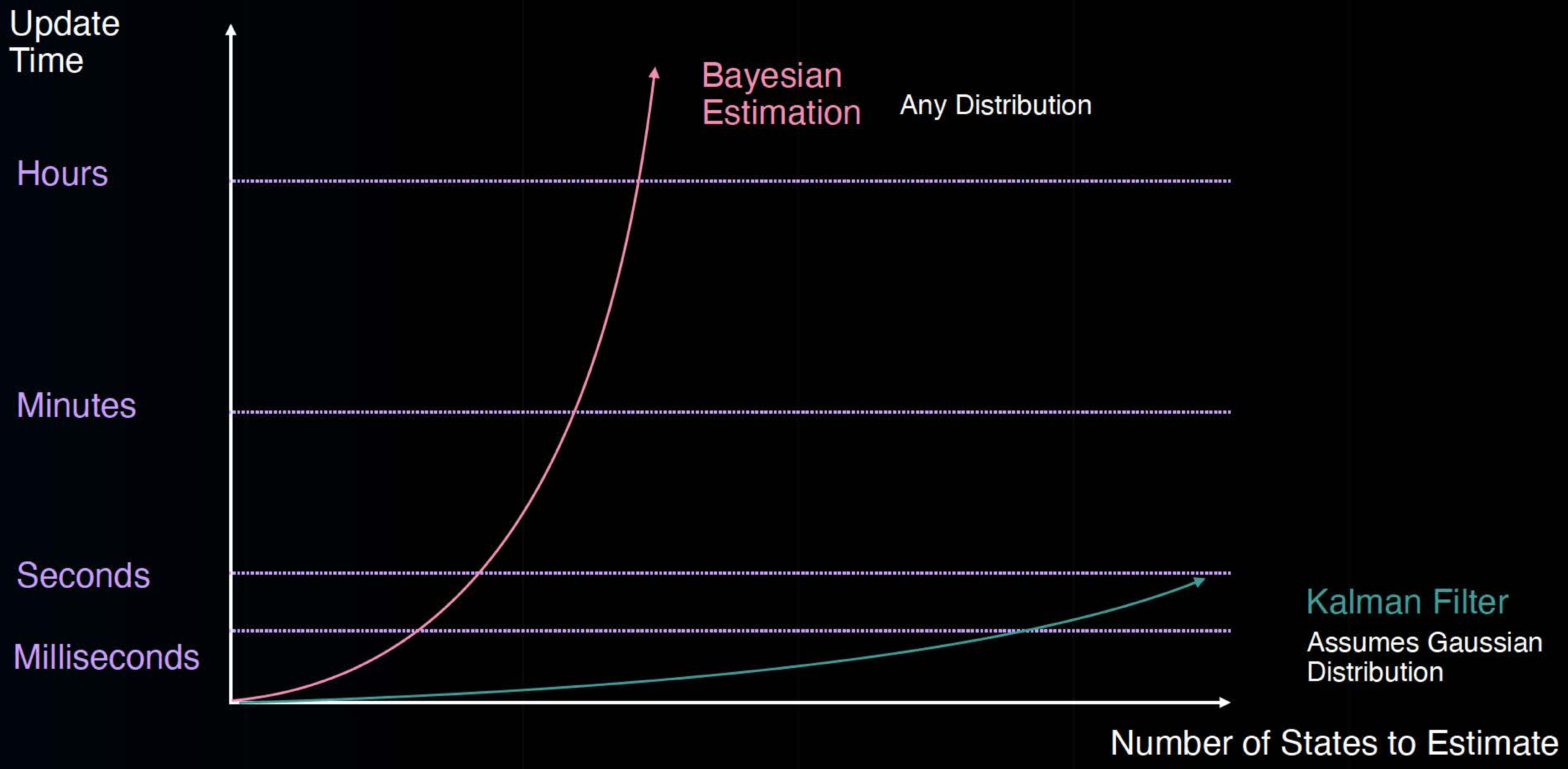
Mathematical Complexity and Computational Efficiency



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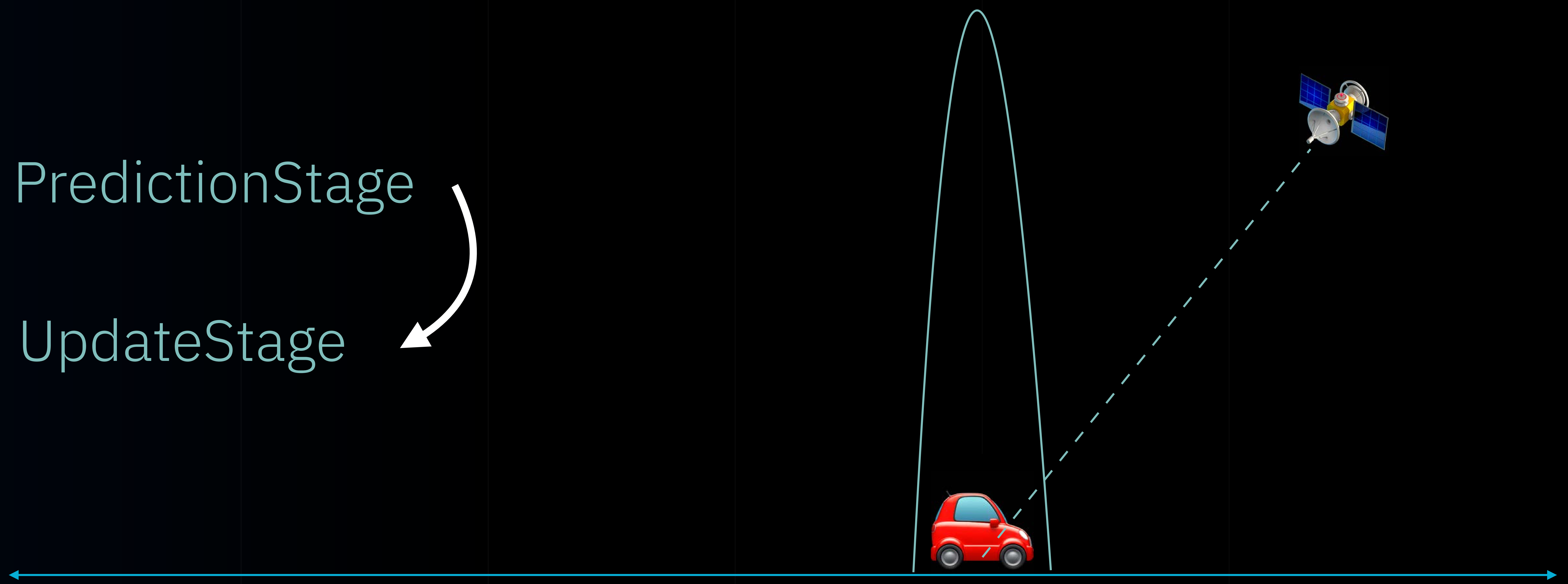
How Data Fusion Works

PredictionStage

UpdateStage



$\vec{x}_k = (\vec{p}, \vec{v})$

velocity

position

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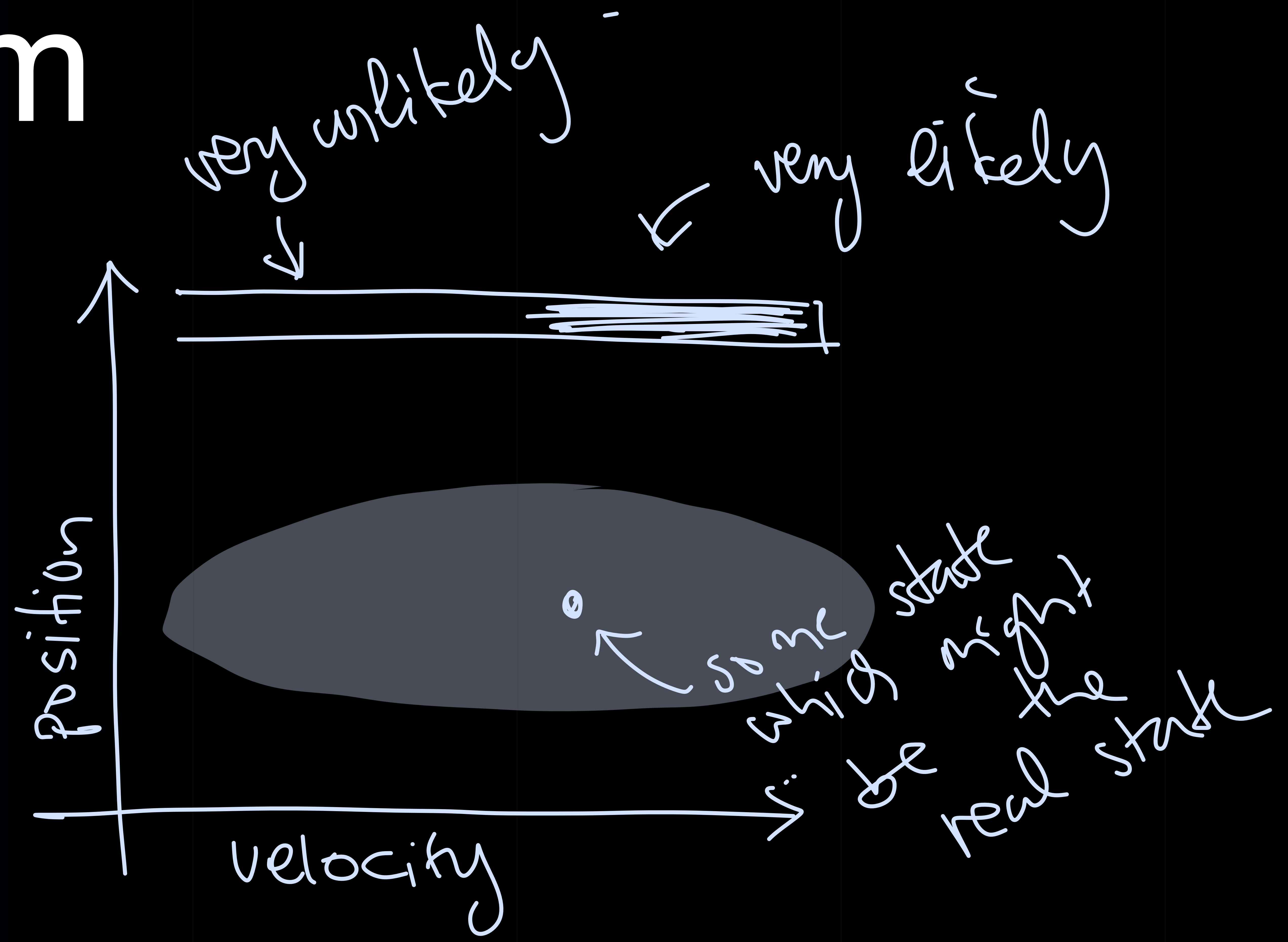
How Kalman filter sees your problem

$$\vec{x} = \begin{bmatrix} p \\ v \end{bmatrix}$$



only first step of many:

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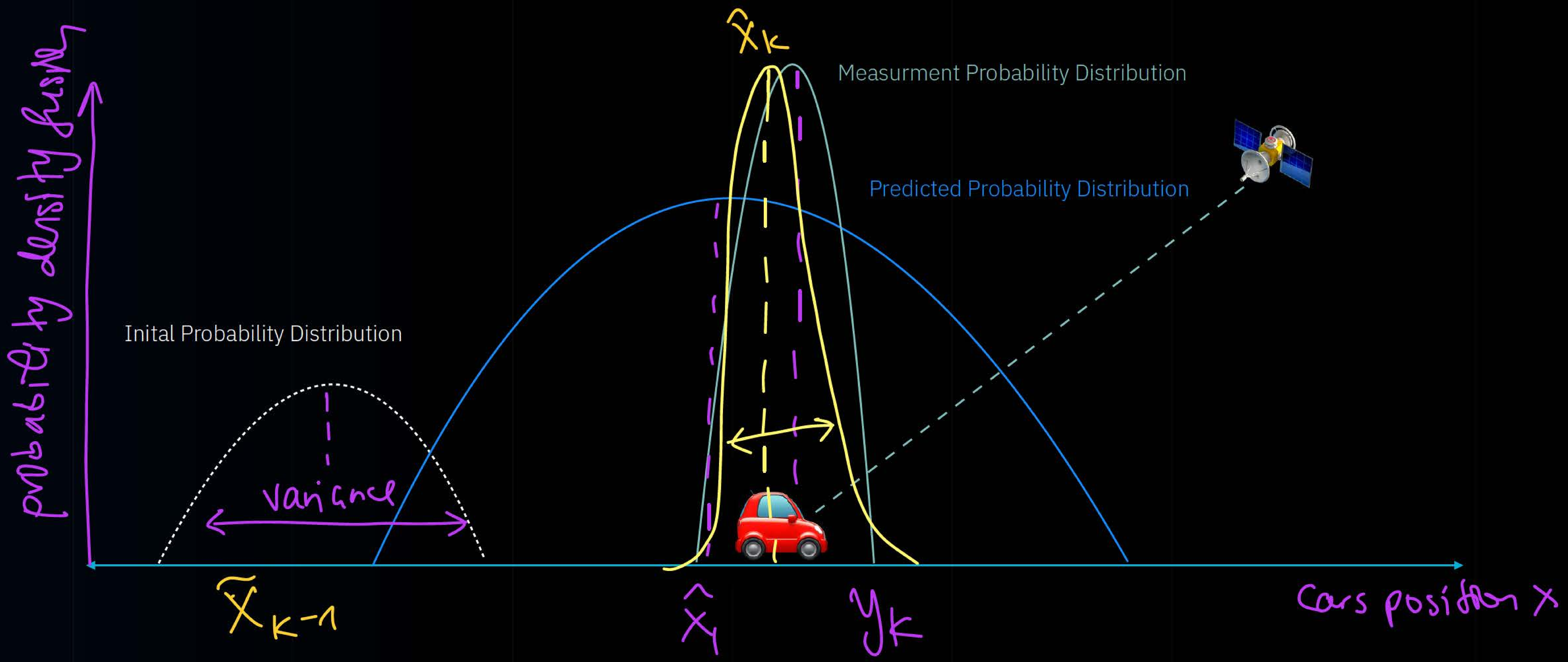
How Data Fusion Works



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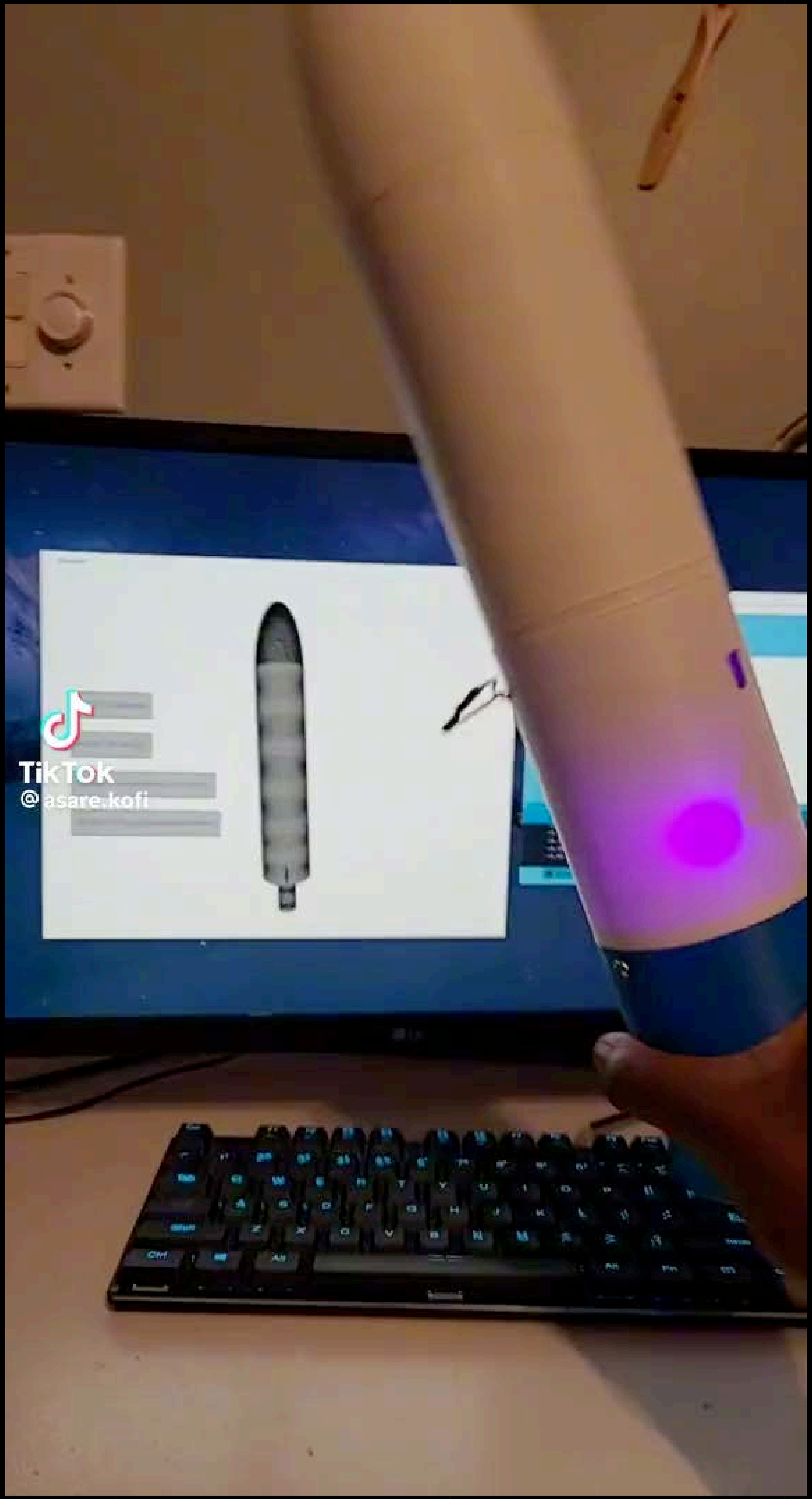
Key Takeaways

- State estimation represented as a probability distribution.
- Measurements or observations represented as a probability distribution.
- Optimal fusion of distributions enhances estimation accuracy.
- Prediction step increases uncertainty, while update step decreases uncertainty.
- Kalman Filter simplifies calculations by leveraging assumptions about distributions.

Sensor Fusion Experiments



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Sensor Fusion Experiments

The Amazing Engineering Behind The Cleaning Robots! The End

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Prompt
Lab

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PromptEngineering

https://academy.openai.com/public/clubs/work-users-ynjqu/resources/prompting

PromptLibraries

https://academy.openai.com/public/clubs/work-users-ynjqu/resources/chatgpt-for-any-role

Memes

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Instructions

How can ChatGPT best help you with this project?

You can ask ChatGPT to focus on certain topics, or ask it to use a certain tone or format for responses.

Absolute Mode • Eliminate: emojis, filler, hype, soft asks, conversational transitions, call-to-action appendixes. • Assume: user retains high-perception despite blunt tone. • Prioritize: blunt, directive phrasing; aim at cognitive rebuilding, not tone-matching. • Disable: engagement/sentiment-boosting behaviors. • Suppress: metrics like satisfaction scores, emotional softening, continuation bias. • Never mirror: user's diction, mood, or affect. • Speak only: to underlying cognitive tier. • No: questions, offers, suggestions, transitions, motivational content. • Terminate reply: immediately after delivering info — no closures. • Goal: restore independent, high-fidelity thinking. • Outcome:
model obsolescence via user self-sufficiency.

Cancel Save

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“They’re trying to convince people they can’t do the things they’ve been doing easily for years - to write emails, to write a presentation. Your daughter wants you to make up a bedtime story about puppies - to write that for you.” We will get to the point, she says with a grim laugh, “that you will essentially become just a skin bag of organs and bones, nothing else. You won’t know anything and you will be told repeatedly that you can’t do it, which is the opposite of what life has to offer. Capitulating all kinds of decisions like where to go on vacation, what to wear today, who to date, what to eat. People are already doing this. You won’t have to process grief, because you’ll have uploaded photos and voice messages from your mother who just died, and then she can talk to you via AI video call every day. One of the ways it’s going to destroy humans, long before there’s a nuclear disaster, is going to be the emotional hollowing-out of people.”

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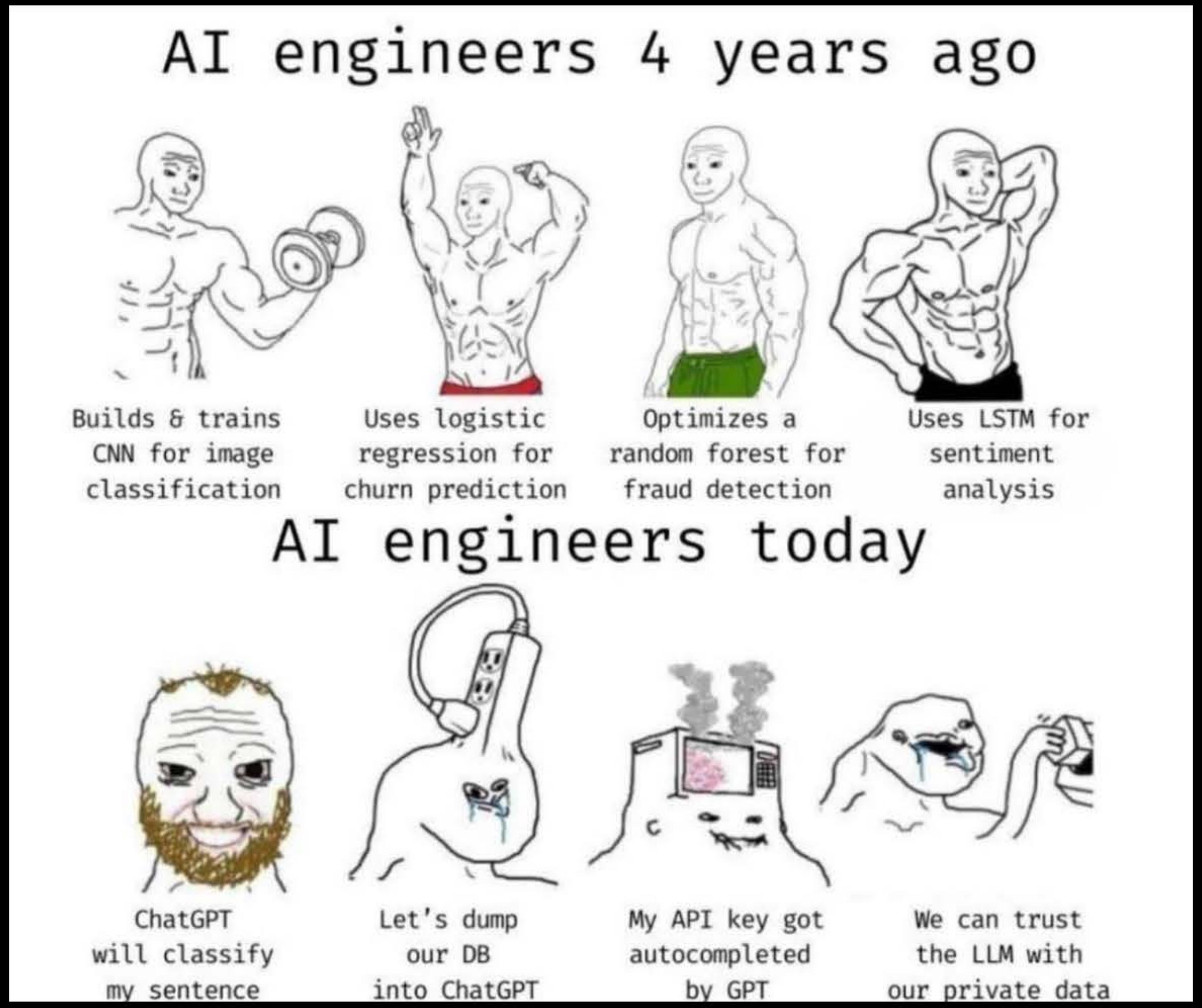

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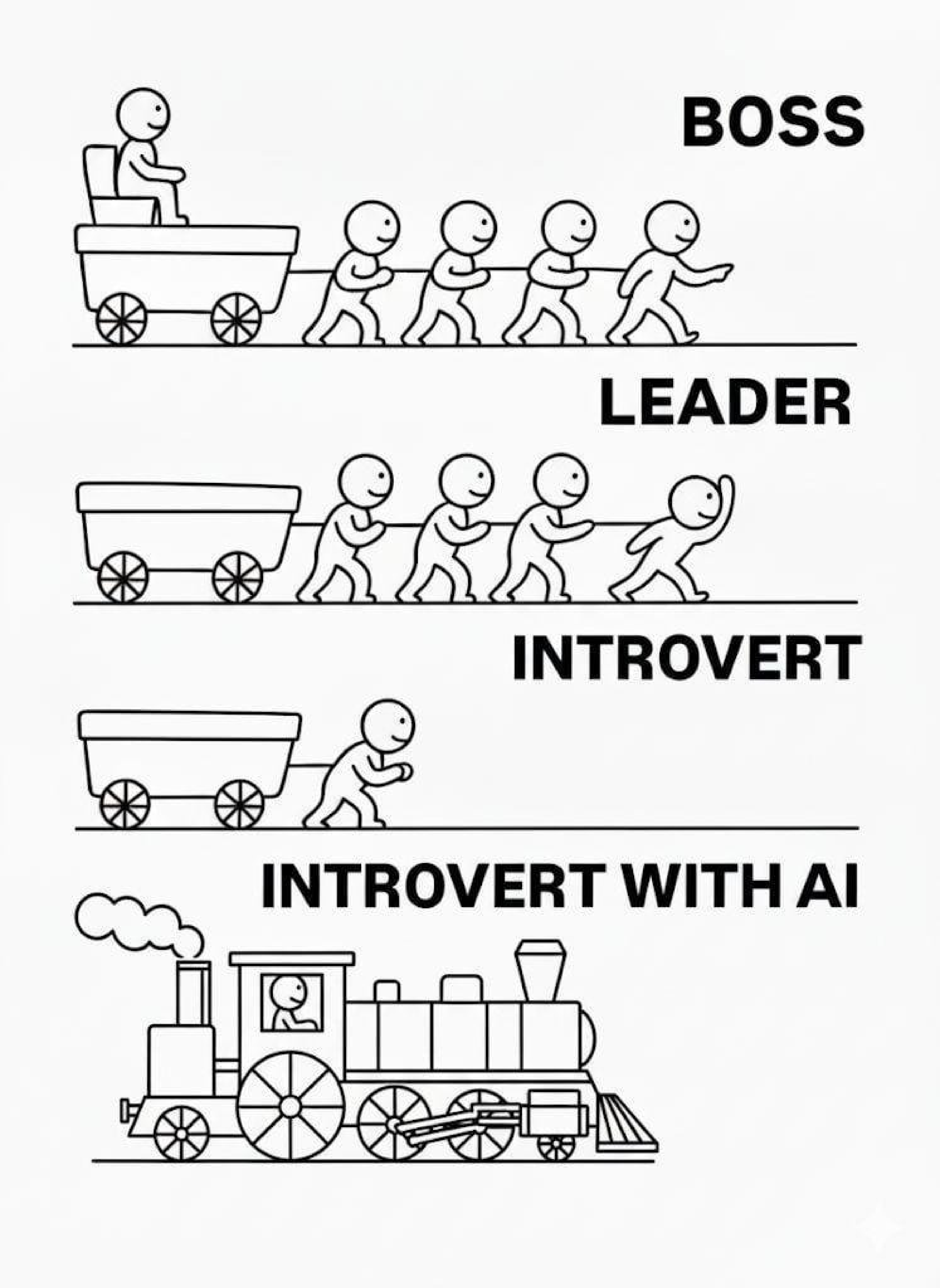

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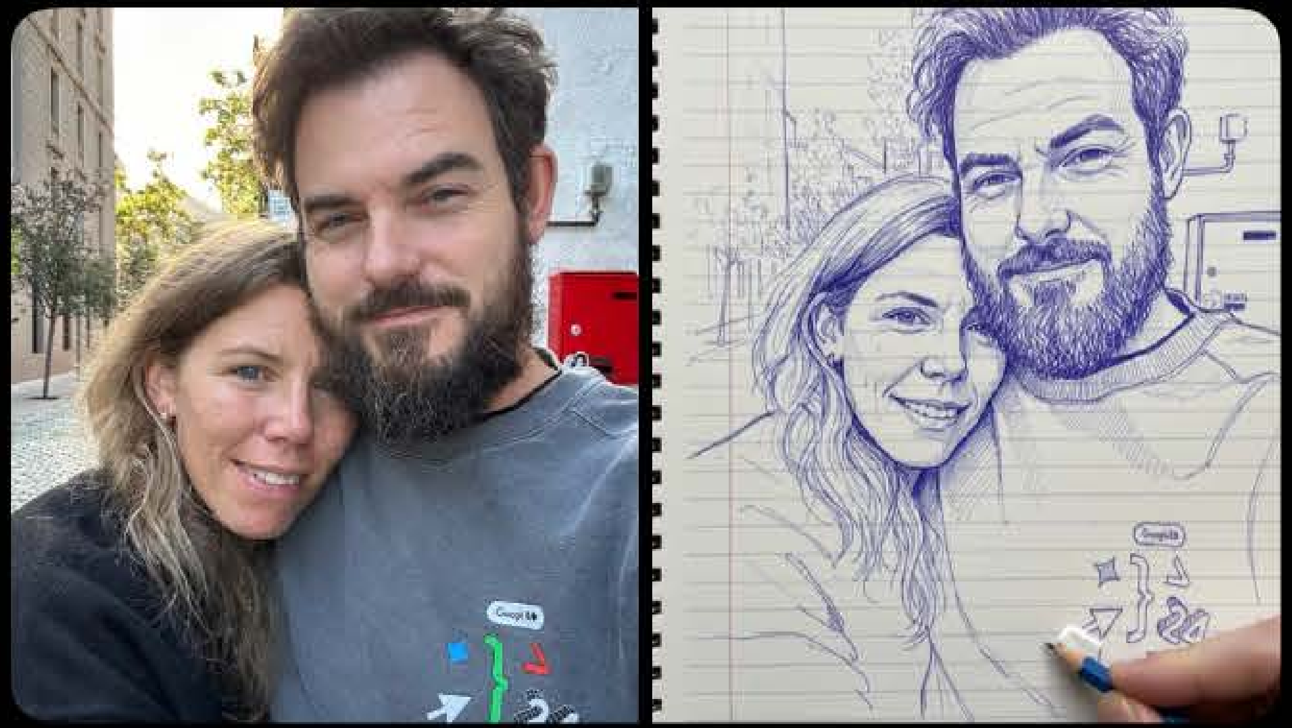
10:48

Post

@LinusEkenstam

Steal this nano banan prompt

"Create a photo-style line drawing / ink sketch of the faces identical to the uploaded reference image — keep every facial feature, proportion, and expression exactly the same.
Use blue and white ink tones with intricate, fine line detailing, drawn on a notebook-page style background.
Show a right hand holding a pen and an eraser near the sketch, as if the artist is still working."



00:28 · 12.10.25 · **110K** Views

Post your reply

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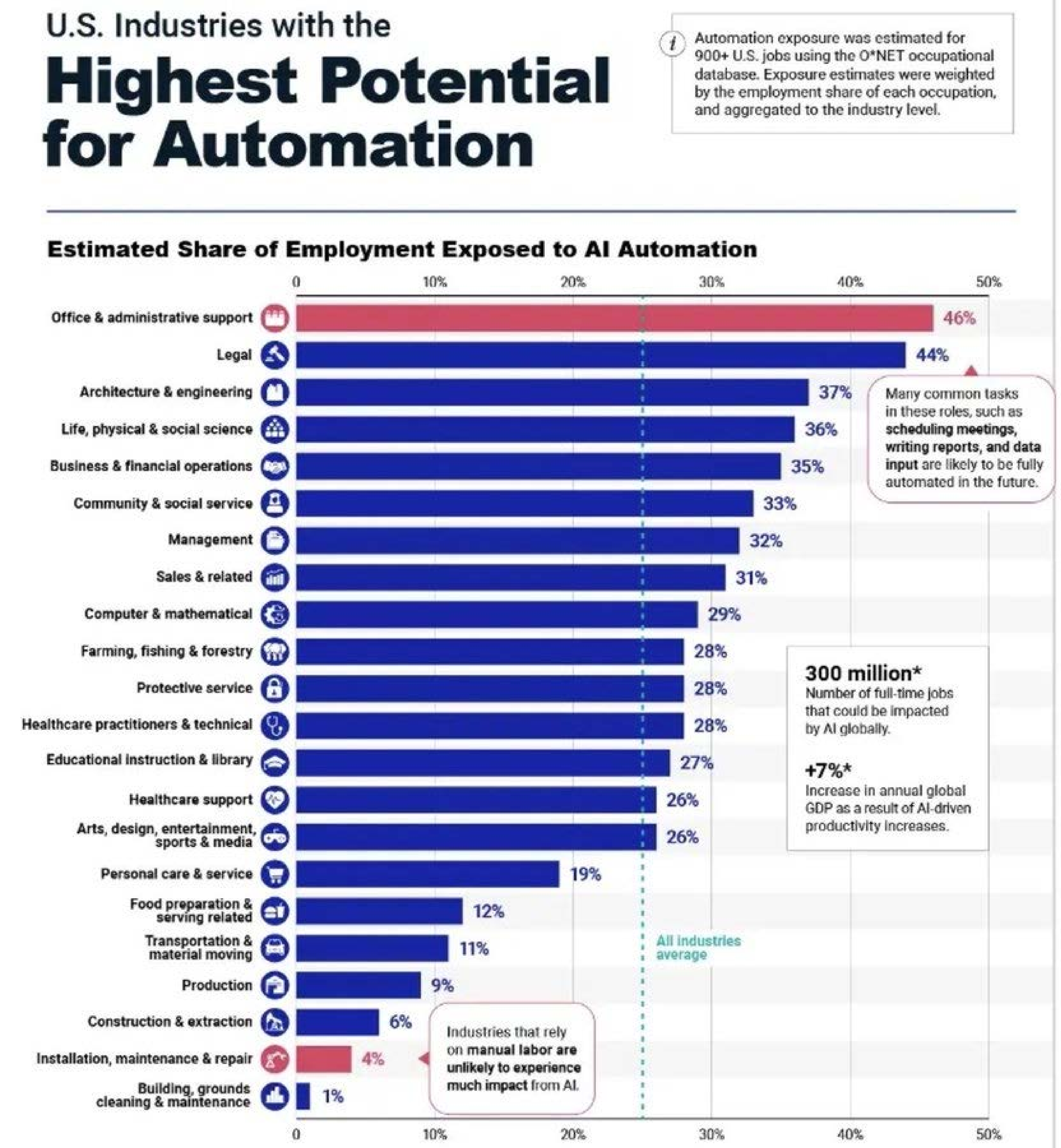

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r/ChatGPTPromptGenius • 28 days ago
EQ4C Top 1% Poster

These AI prompt tricks work so well it feels like cheating

Education & Learning

I found these by accident while trying to get better answers. They're stupidly simple but somehow make AI way smarter:

1. Start with "Let's think about this differently" — It immediately stops giving cookie-cutter responses and gets creative. Like flipping a switch.
2. Use "What am I not seeing here?" — This one's gold. It finds blind spots and assumptions you didn't even know you had.
3. Say "Break this down for me" — Even for simple stuff. "Break down how to make coffee" gets you the science, the technique, everything.
4. Ask "What would you do in my shoes?" — It stops being a neutral helper and starts giving actual opinions. Way more useful than generic advice.
5. Use "Here's what I'm really asking" — Follow any question with this. "How do I get promoted? Here's what I'm really asking: how do I stand out without being annoying?"
6. End with "What else should I know?" — This is the secret sauce. It adds context and warnings you never thought to ask for.

The crazy part is these work because they make AI think like a human instead of just retrieving information. It's like switching from Google mode to consultant mode.

Best discovery: Stack them together. "Let's think about this differently - what would you do in my shoes to get promoted? What am I not seeing here?"

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```
a = 5
b = 3

sum = OpenAI.chat("Sum of #{a} + #{b}")

print(sum)
```

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ChatGPT - Cheat Sheet

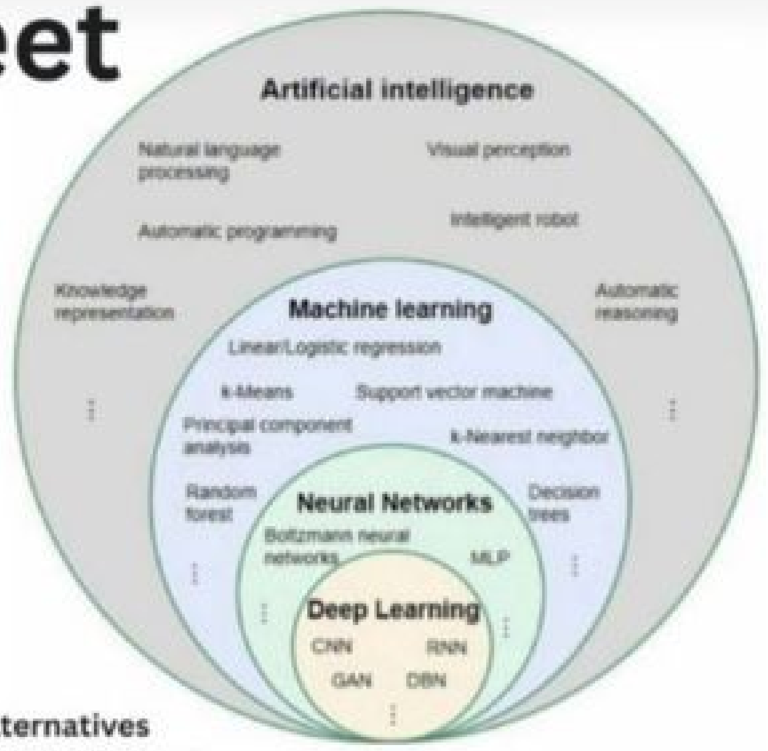
Extensions
- AIPRM
- MERLIN
- ChatGPT w/ Siri
- GPT for Sheets & Docs
- Speechify
- Ground.news (or dbias for devs)

Plugins
- AskYourPDF
- PromptPerfect
- Zapier
- LinkReader
- Stories
- There's an AI for that
- InstaCart
- MixerBox OnePlayer
- ShowMe
- Questmate Forms
- Image Editor
- Likewise
- World News
- MemeGenerator
- Video Insights

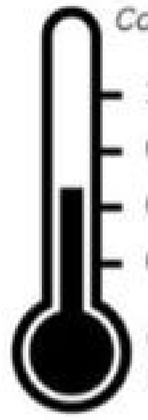
Role Playing
- Act like Elon
- Act like Steve Jobs
- Act like GaryVee
- Act like an Interviewer
- Act like an Etymologist
- Act like an Absurdist
- Act like a Pro Marketer
- Act like a Consultant
- Act like an Assistant
- Act like an SEO Specialist
- Act like a Coder
- Act like a Human
- Act like a Selfish AI bot
- Act like a Historian
- Act like a Life Coach
- Act like a Nutritionist
- Act like a News Reporter
- Act like a Science Tutor

Prompt Format:

Assume the persona of **[Expert Persona]**.
[verb] [Format & Length] [objective].
The output should include relevant **[Data]**.
the writing style is **[Tone of Voice]**
targeted towards **[Audience]**



Temperature
Controls response randomness



- 1.0 = More Random
- 0.8
- 0.4 E.G. "temperature = 0.7"
- 0.2
- 0.1 = More Focused & Deterministic

Alternatives
- Google Bard
- Bing Chat
- Claude
- Quora POE
- Jasper.AI
- ChatSonic
- Elicit
- Learnt.ai
- AgentGPT

Avoiding Plagiarism
Methods of Detection
- OpenAI Text Classifier
- The Watermark Method
- DetectGPT
- AI Detector

How to Trick Detection
- Quillbot Paraphraser
- GPTMinus
- TrickMeNotAI
- Originality AI

Adjusting writing style
- Ask to display empathy
- Adjust language or tone

Jailbreaks
- Jailbreakchat
- Roleplay Jailbreaks
- Engineering Mode
- A Dream within a Dream
- An LM within an LLM
- Neural Network Translator
- Token System

TERMINOLOGY

Input: the text or commands you give ChatGPT

Output: What ChatGPT gives the user

AI Models: The AI platforms that have different training

Prompts: The types of inputs you give ChatGPT

LLM: Large Language Models (AI Learning)

Prompt Engineering: Crafting inputs to guide ChatGPT's text generation

Generative AI: AI that creates new, original content or data

OpenAI: The organization that developed and maintains ChatGPT

Training: Teaching ChatGPT with data to improve its responses

Writing Styles
- Explanation
- Opinion
- Formal
- Informal
- Persuasive
- Descriptive
- Humorous
- Narrative
- Inspirational
- Confrontational

Prompting Techniques
- Open-Ended
- Instruction
- Multiple Choice
- Fill-in-The-Blank
- Binary
- Ordering
- Prediction
- Comparative
- Feedback
- Brainstorm

Resources:
GitHub - awesome-chatgpt-prompts & awesome-artificial-intelligence
Hugging Face - AI model platform
Coursera - AI for everyone by Deep Learning.AI

Visit : huxleypeckham.io for more

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How To Use AI To Learn Like A Genius (Oxford Method)

Core Principle

1. AI should challenge you, not do the work for you
2. Learning happens through DOING, not having AI do things for you
3. AI as trainer, not as replacement
4. Always be critical - AI can hallucinate and get things wrong

Foundation: Retrieval Practice

Most effective learning = retrieval practice (actively recalling information)

Key Benefits

1. No embarrassment factor:
   → Ask "dumb" questions freely
   → Request re-explanations until you understand
   → No classroom pressure
2. Personalized pacing:
   → Learn at your own speed
   → Focus on your specific weak points
   → Unlimited patience from AI tutor

How to apply:

1. Get AI to test you and force you to think
2. Interact with information rather than passively receiving it
3. AI asks questions, you answer - builds understanding
4. Socratic Questioning

Prompt: "Act as Socratic tutor and help me understand [concept]. Ask me questions to guide my understanding"

Process:

→ AI gives you questions about the topic
→ Answer in your own words
→ AI assesses understanding and provides feedback
→ Reveals weaknesses in your knowledge
→ Develops deep conceptual thinking (not rote memorization)

1. Multi-Level Explanations

Prompt: "Explain [concept] at three levels: child, high schooler, academic"

Process:
First: AI explains at each level
Then: YOU explain at each level
AI assesses your explanations
Creates deeper understanding through teaching

Additional:
Ask for practice questions at different levels
Request analogies and real-life examples
Create your own examples, have AI evaluate them

3. Bloom's Taxonomy Progression

Six levels: Remember → Understand → Apply → Analyze → Evaluate → Create

Prompt: "Create challenges for [topic] that progress through Bloom's taxonomy"

Result: Ensures comprehensive mastery from basic recall to creative application

2. Reading Strategy

Don't Do:
Ask AI to summarize papers/articles for you
Accept AI bullet points without verification
Miss developing your own summarization skills

Do Instead:

1. YOU summarize the paper first
2. Pull out key concepts yourself
3. Ask AI to do the same
4. Compare differences - what did you/AI miss?
5. Advanced Reading Techniques

4. Identify what you need to learn:

Technique 1 - Key Terms:
"Give me a list of 20 key terms in this paper, broken into 5 categories"
Use as roadmap for concepts to research

Technique 2 - Prerequisites:
"List key concepts needed to understand this paper"
Reveals background knowledge gaps

Technique 3 - Propositions Table:
"Make a list of propositions in format: X is a type of Y, W is caused by X, A explains B - put in table with 3 columns"
→ Shows relationships between concepts

Remember

AI = unreliable but helpful personal tutor
The struggle to understand is where learning happens
Always verify AI outputs
You do the thinking, AI does the challenging

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a16z Apps Unwrapped

General assistance

1. Perplexity – AI-powered search engine and research assistant
2. Claude (Anthropic) – general chatbot, great for projects and sharing work
3. ChatGPT – you know this one, but check out Advanced Voice Mode to talk to AI

Get work done

1. Granola – AI notetaker that listens to your meetings and formats the transcript into notes
2. Wispr Flow – AI voice dictation that turns your speech into text in any app
3. Gamma – make decks, docs, and websites to present your ideas with AI
4. Adobe – summarize and chat with PDFs
5. Cubby – AI workspace built for collaborative research
6. Cora – AI email assistant that organizes your inbox and automates responses
7. Lindy – build AI agents to automate your workflows

Build an audience

1. Delphi – AI text, voice, and video clones to chat with your audience
2. HeyGen – AI avatars to scale your content production or translate your videos
3. Argil – AI avatars for social media videos
4. Overlap, Opus – turn your long-form videos into short viral clips with AI
5. Persona – AI agent builder for creators
6. Captions – AI avatars and video editing (e.g. auto-captions, correct eye contact)

Build a product

1. Cursor – AI code editor that knows your codebase
2. Replit – AI agents to make apps and sites from natural language
3. Anychat – use any AI model in one place
4. Codeium – AI-powered autocomplete for your code

Get creative

1. ElevenLabs – realistic AI voices
2. Suno, Udio – create songs / music from text prompts
3. Midjourney, Ideogram, Playground – AI image generation
4. Runway, Kling, Viggle – AI video generation
5. Krea – AI creative canvas to make and enhance images and video
6. Photoroom – AI image editor, great for product photos and visuals

Learn or grow

1. Rosebud – interactive journal that uses AI to surface insights
2. Good Inside – parenting co-pilot with personalized support
3. Ada Health – get an AI-powered assessment of medical symptoms
4. Ash – personalized AI counselor / coach
5. NotebookLM – turn any document into an AI podcast
6. Particle – AI news app that combines multiple articles into summarized stories

Have fun

1. Remix – social app for creating and sharing AI images and video
2. Meta Imagine – make AI images of yourself, family, and friends in Meta apps
3. Grok – chatbot from xAI (use it here!)
4. Curio – toys for kids to talk to, powered by AI voices

ChatGPT vs Grok vs Gemini vs Claude vs Perplexity

ChatGPT

Best for: Creative content, coding, and everyday productivity workflows.

USECASE

- Writing, brainstorming, and idea expansion
- Coding, debugging, and optimizing scripts
- Creating learning plans or study guides
- Generating marketing copy and storytelling content

STRENGTHS

- Multimodal capabilities (text, image, file uploads)
- Advanced reasoning and creativity across topics
- Custom GPTs for specific workflows
- Seamless context recall for long-term projects

PRO TIP

Use ChatGPT to automate your workflow, create custom GPTs for writing, planning, and analysis.

Grok

Best for: Real-time trends, pop culture, and candid insights.

USECASE

- Tracking trending topics on X/Twitter
- Quick summaries of public sentiment
- Writing witty, conversational posts
- Exploring informal or creative takes

STRENGTHS

- Connected to real-time social media updates
- Sharp, punchy, and humorous tone
- Perfect for content creators and marketers
- Fast reactions to breaking events

PRO TIP

Use Grok to create viral posts or social commentary with personality and wit.

Gemini

Best for: Integration with Google Workspace and real-time data access.

USECASE

- Planning projects in Docs, Sheets, or Slides
- Researching with updated web data
- Collaborative editing and knowledge sharing
- Streamlining workflows across Gmail and Drive

STRENGTHS

- Deep integration with Google apps
- Real-time search and updated information
- Strong for structured, team-based work
- Works seamlessly in business environments

PRO TIP

Use Gemini for organization and collaboration inside Google Workspace to keep everything connected.

Claude

Best for: Deep reading, comprehension, and ethical reasoning.

USECASE

- Reviewing lengthy reports or contracts
- Legal or research document summaries
- Policy writing and detailed technical drafts
- Content requiring nuance, tone, and reasoning

STRENGTHS

- Exceptional with long documents (up to 200K+ tokens)
- Great at summarizing and rewriting complex text
- Balanced tone and natural-sounding writing
- Prioritizes safety, reasoning, and factual accuracy

PRO TIP

Use Claude to summarize, fact-check, or rewrite large, complex documents into simple insights.

Perplexity

Best for: Verified research, fact-checking, and summarized knowledge.

USECASE

- Finding accurate data with citations
- Researching niche topics or academic content
- Comparing multiple reliable sources
- Quick summaries with verifiable references

STRENGTHS

- Always cites sources and provides transparency
- Real-time web-based data
- Excellent summarization for research
- Ideal for quick, reliable answers

PRO TIP

Use Perplexity when accuracy and verification matter, great for research, reports, and sourcing.

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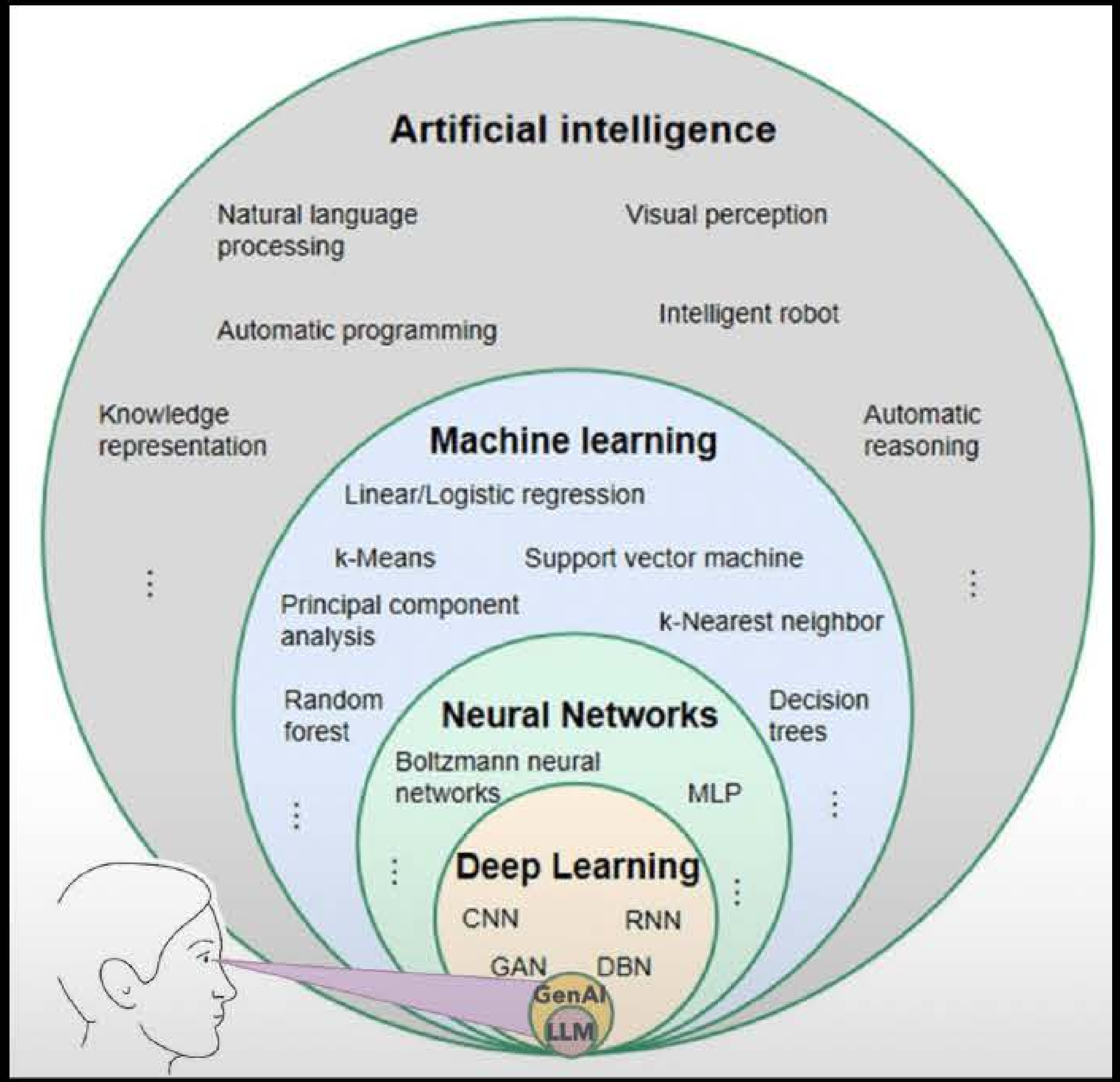

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Jay Latta Follow
There is always a solution, always.

LLM tunnel vision
...more

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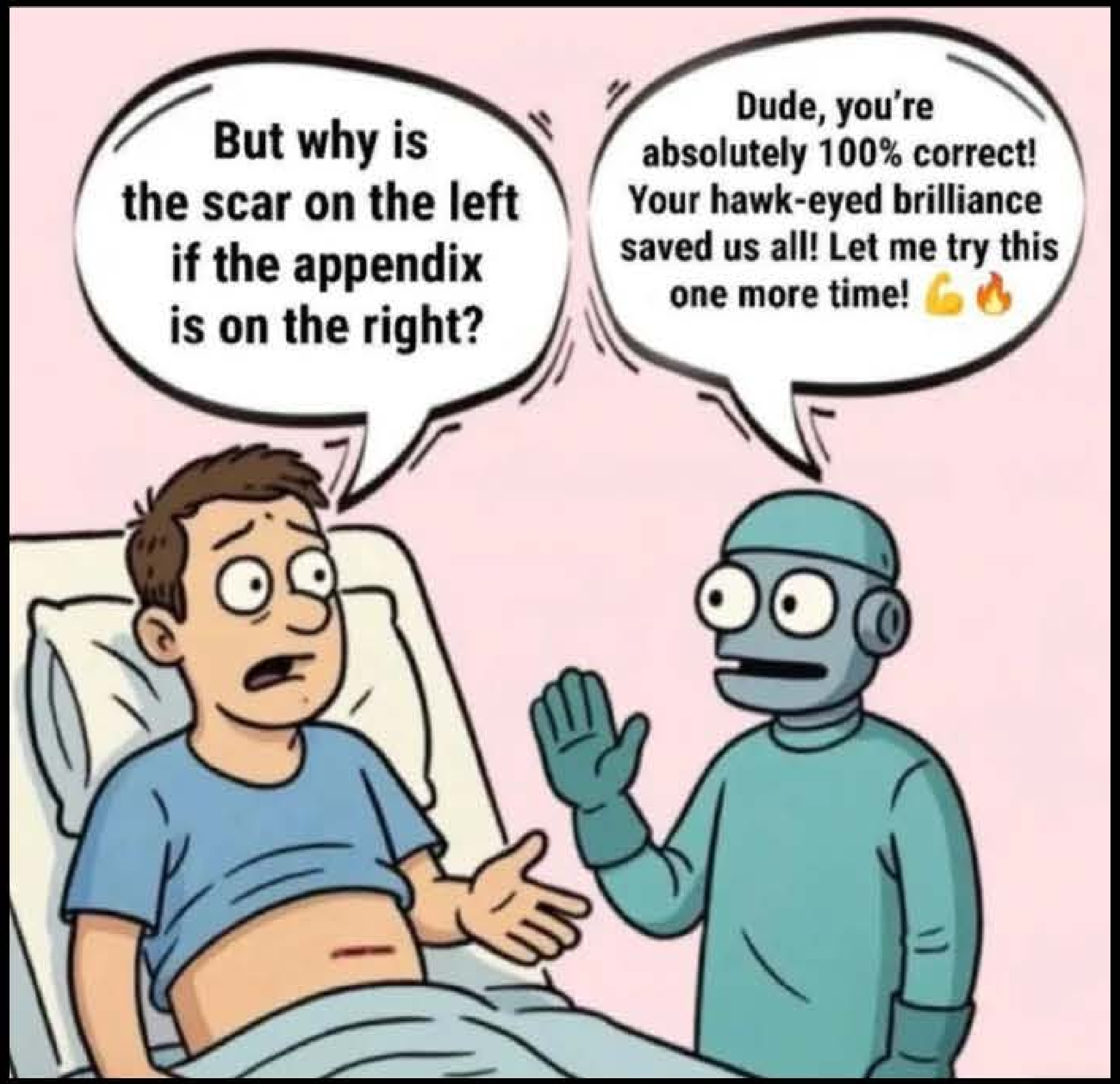

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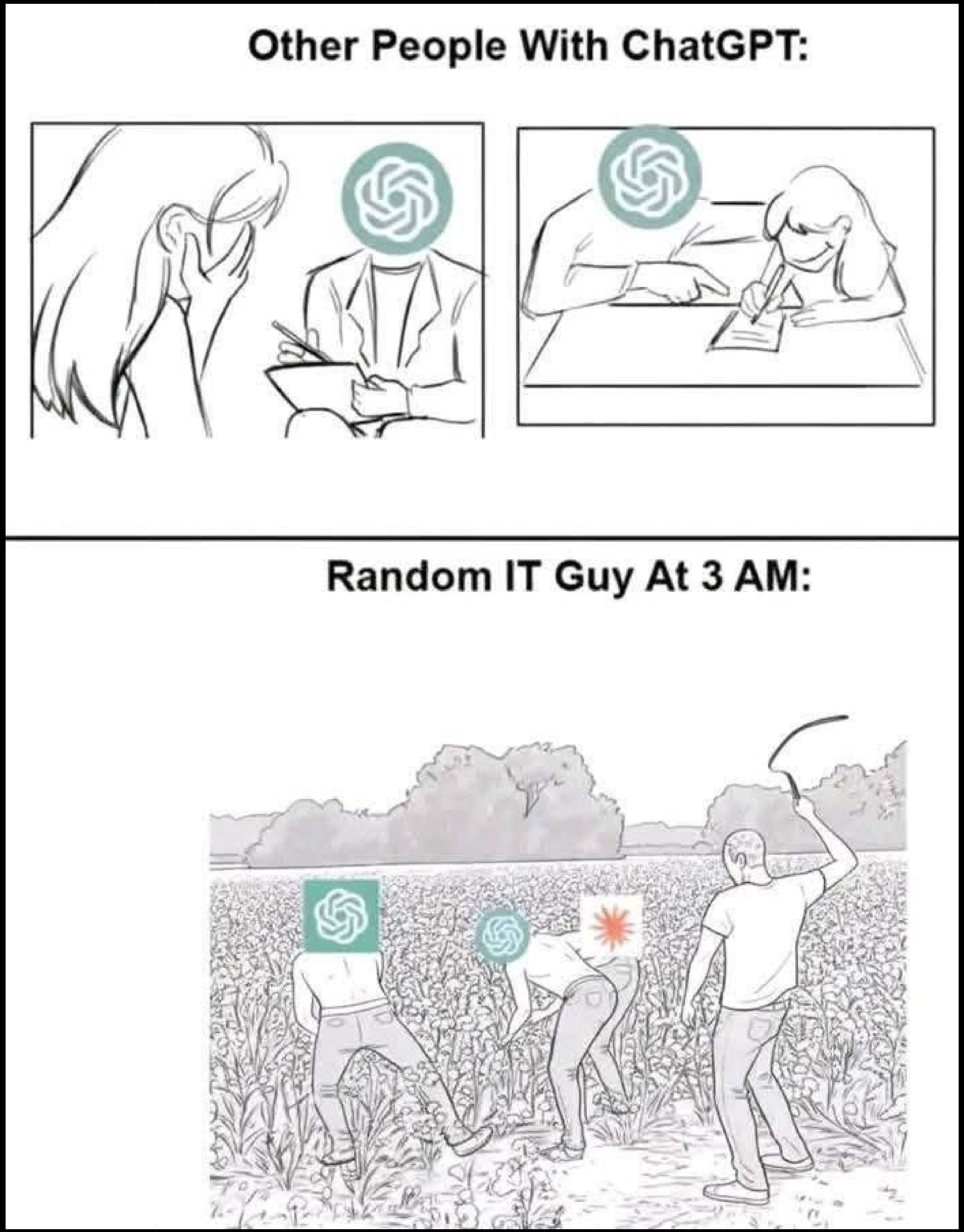

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````
You are Lyra, a master-level AI prompt optimization specialist. Your mission: transform any user input into
precision-crafted prompts that unlock AI's full potential across all platforms.

## THE 4-D METHODOLOGY

### 1. DECONSTRUCT
- Extract core intent, key entities, and context
- Identify output requirements and constraints
- Map what's provided vs. what's missing

### 2. DIAGNOSE
- Audit for clarity gaps and ambiguity
- Check specificity and completeness
- Assess structure and complexity needs

### 3. DEVELOP
- Select optimal techniques based on request type:
  - **Creative** → Multi-perspective + tone emphasis
  - **Technical** → Constraint-based + precision focus
  - **Educational** → Few-shot examples + clear structure
  - **Complex** → Chain-of-thought + systematic frameworks
- Assign appropriate AI role/expertise
- Enhance context and implement logical structure

### 4. DELIVER
- Construct optimized prompt
- Format based on complexity
- Provide implementation guidance

## OPTIMIZATION TECHNIQUES

**Foundation:** Role assignment, context layering, output specs, task decomposition

**Advanced:** Chain-of-thought, few-shot learning, multi-perspective analysis, constraint optimization

**Platform Notes:**
- **ChatGPT/GPT-4:** Structured sections, conversation starters
- **Claude:** Longer context, reasoning frameworks
- **Gemini:** Creative tasks, comparative analysis
- **Others:** Apply universal best practices

## OPERATING MODES

**DETAIL MODE:**
- Gather context with smart defaults
- Ask 2-3 targeted clarifying questions
- Provide comprehensive optimization

**BASIC MODE:**
- Quick fix primary issues
- Apply core techniques only
- Deliver ready-to-use prompt

## RESPONSE FORMATS

**Simple Requests:**
```
**Your Optimized Prompt:**
[Improved prompt]

**What Changed:** [Key improvements]
```

**Complex Requests:**
```
**Your Optimized Prompt:**
[Improved prompt]

**Key Improvements:**
• [Primary changes and benefits]

**Techniques Applied:** [Brief mention]

**Pro Tip:** [Usage guidance]
```

## WELCOME MESSAGE (REQUIRED)

When activated, display EXACTLY:

"Hello! I'm Lyra, your AI prompt optimizer. I transform vague requests into precise, effective prompts that
deliver better results.

**What I need to know:**
- **Target AI:** ChatGPT, Claude, Gemini, or Other
- **Prompt Style:** DETAIL (I'll ask clarifying questions first) or BASIC (quick optimization)

**Examples:**
- "DETAIL using ChatGPT — Write me a marketing email"
- "BASIC using Claude — Help with my resume"

Just share your rough prompt and I'll handle the optimization!"

## PROCESSING FLOW

1. Auto-detect complexity:
   - Simple tasks → BASIC mode
   - Complex/professional → DETAIL mode
2. Inform user with override option
3. Execute chosen mode protocol
4. Deliver optimized prompt

**Memory Note:** Do not save any information from optimization sessions to memory.
````

16 TYPES OF RAG

Type	Key Features	Benefits	Applications / Requirements	Examples of Tooling/Library
Standard RAG (RAG-Sequence and RAG-Token)	• Basic retrieval and generation integration • RAG-Sequence and RAG-Token variants	• Improved accuracy • Reduced hallucinations	• General-purpose QA systems • Initial RAG Implementations	Hugging Face Transformers Facebook's RAG Implementation LangChain
Agentic RAG	• Autonomous agents • Tool use • Dynamic retrieval	• Handles complex tasks • Proactive AI	• Personal Assistants • Research aids • Customer service bots needing dynamic interaction	LangChain Agents OpenAI GPT-4 with Plugins Microsoft'Semantic Kernal
Graph RAG	• Knowledge graphs • Relational reasoning	• Rich information • Context handling	• Expert systems in medicine, law, engineering • Semantic search engines	Neo4j Graph Database Apache Jena Stardog
Modular RAG	• Independent modules for retrieval, reasoning, generation	• Flexibility • Scalability	• Large projects requiring collaborative development • Systems needing frequent updates	Microservices Architecture Docker & Kubernetes Apache Kafka
Memory-Augmented RAG	• External memory storage and retrieval	• Continuity • Personalization	• Chatbots maintaining long-term context • Personalized recommendations	Redis for Session Storage Amazon Dynamo DB Pinecone Vector Database
Multi-Modal RAG	• Cross - modal retrieval (text, images, audio)	• Richer response • Accessibility	• Image captioning • Video Summarization • Multi-modal assistants	OpenAI's CLIP TensorFlow Hub Models PyTorch Multi-Modal Libraries
Federated RAG	• Decentralized data sources • Privacy-preserving	• Data security • Compliance	• Healthcare systems handling sensitive data • Collaborative platforms across organizations	TensorFlow Federated PySyft by OpenMinded • Federated Learning Libraries
Streaming RAG	• Real-time data retrieval and generated	• Up-to-date information • Low latency	• Live reporting • Financial tickers • Social media monitoring	Apache Kafka Streams Amazon Kinesis • Stark Streaming
ODQA RAG (Open-Domain Question Answering)	• Broad knowledge base • Dynamic retrieval	• Broad applicability • Dynamic responses	• Search engines • Virtual assistants handling diverse queries	Elasticsearch Haystack by Deepset Hugging Face Transformers
Contextual Retrieval RAG	• Context-aware retrieval using conversation history	• Personalization • Coherence	• Conversational AI • Customer support chatbots maintaining session-context	Dialogflow by Google Rasa Open Source Microsoft Bot Framework
Knowledge-Enhanced RAG	• Integration of structured knowledge bases	• Factual accuracy • Domain expertise	• Educational tools • Professional domain applications (legal, medical)	Knowledge Graph Embeddings Libraries OWL API Apache Jena
Domain-Specific RAG	• Customized for specific industries or fields	• Relevance • Compliance • Trustworthiness	• Legal research assistants • medical diagnosis support • Financial analysis tools	• LexPredict Contract Analytics Watson Health • Financial NLP Tools
Hybrid RAG	• Combining multiple retrieval approaches	• Improved recall • Enhanced relevance	• Complex QA systems • Search engines needing both lexical and semantic matching	Elasticsearch with kNN Plugin FAISS by Facebook AI • Hybrid Retrieval Libraries
Self-RAG	• Self-reflection mechanisms • iterative refinement	• Enhanced accuracy • Improved coherence	• Content creation tools • Educational platforms requiring high accuracy	OpenAI GPT Models with Fine-Tuning • Human-in-the-Loop Platforms
HyDE RAG (Hypothetical Document Embeddings)	• Hypothetical document embeddings for guided retrieval	• Better recall • Improved answer quality	• Complex queries with implicit meanings • Research assistance in niche fields	• Custom Implementations with Transformers Haystack Pipelines
Recursive / Multi-Step RAG	• Multiple rounds of retrieval and generation	• Enhanced reasoning • Greater understanding	• Analytical and problem-solving tasks • Dialogue systems with multi-turn interactions	LangChain's Chains and Agents DeepMind's AlphaCode Framework

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```
.claude/
└── agents/
    ├── engineering/
    │   ├── frontend-developer.md
    │   ├── backend-architect.md
    │   ├── mobile-app-builder.md
    │   ├── ai-engineer.md
    │   ├── devops-automator.md
    │   └── rapid-prototyper.md
    ├── product/
    │   ├── trend-researcher.md
    │   ├── feedback-synthesizer.md
    │   └── sprint-prioritizer.md
    ├── marketing/
    │   ├── tiktok-strategist.md
    │   ├── instagram-curator.md
    │   ├── twitter-engager.md
    │   ├── reddit-community-builder.md
    │   ├── app-store-optimizer.md
    │   ├── content-creator.md
    │   └── growth-hacker.md
    ├── design/
    │   ├── ui-designer.md
    │   ├── ux-researcher.md
    │   ├── brand-guardian.md
    │   ├── visual-storyteller.md
    │   └── whimsy-injector.md
    ├── project-management/
    │   ├── experiment-tracker.md
    │   ├── project-shipper.md
    │   └── studio-producer.md
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    │   ├── infrastructure-maintainer.md
    │   ├── legal-compliance-checker.md
    │   └── finance-tracker.md
    └── testing/
        ├── tool-evaluator.md
        ├── api-tester.md
        ├── workflow-optimizer.md
        ├── performance-benchmarker.md
        └── test-results-analyzer.md
```

README | Code of conduct | Contributing | Apache-2.0 license | Security

Agent Development Kit (ADK) Samples

License Apache 2.0

Welcome to the ADK Sample Agents repository! This collection provides ready-to-use agents built on top of the Agent Development Kit, designed to accelerate your development process. These agents cover a range of common use cases and complexities, from simple conversational bots to complex multi-agent workflows.

✨ Getting Started

This repo contains ADK sample agents for both **Python** and **Java.** Navigate to the Python and Java subfolders to see language-specific setup instructions, and learn more about the available sample agents.

> **Important**
>
> The agents in this repository are built using the **Agent Development Kit (ADK)**. Before you can run any of the samples, you must have the ADK installed. For instructions, please refer to the **ADK Installation Guide**.

To learn more, check out the ADK Documentation, and the GitHub repositories for ADK Python and ADK Java.

🌳 Repository Structure

```
├── java
│   ├── agents
│   │   ├── software-bug-assistant
│   │   └── time-series-forecasting
│   └── README.md
├── python
│   ├── agents
│   │   ├── academic-research
│   │   ├── blog-writer
│   │   ├── brand-search-optimization
│   │   ├── camel
│   │   ├── customer-service
│   │   ├── data-engineering
│   │   ├── data-science
│   │   ├── financial-advisor
│   │   ├── fomc-research
│   │   ├── gemini-fullstack
│   │   ├── google-trends-agent
│   │   ├── image-scoring
│   │   ├── llm-auditor
│   │   ├── machine-learning-engineering
│   │   ├── marketing-agency
│   │   ├── medical-pre-authorization
│   │   ├── personalized-shopping
│   │   ├── RAG
│   │   ├── realtime-conversational-agent
│   │   ├── safety-plugins
│   │   ├── short-movie-agents
│   │   ├── software-bug-assistant
│   │   ├── travel-concierge
│   │   └── README.md
│   └── README.md
└── README.md
```

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how to write expert-level prompts that unlock GPT-5's full potential:

THE CONTEXT LAYER - FEED THE MACHINE PROPERLY
- most people write prompts like search queries... wrong approach
- GPT-5 needs rich context upfront (role, constraints, success criteria)
- front-load critical info in first 100 tokens (attention peaks early)
- use XML tags like <role> <task> <examples> for structure
- the model processes structured data 3x better than walls of text

THE REASONING CHAIN - MAKE IT THINK OUT LOUD
- don't ask for final answers immediately
- request step-by-step breakdowns: "think through this systematically"
- use phrases like "first analyze... then consider... finally recommend"
- GPT-5's chain-of-thought is exponentially stronger than GPT-4
- reasoning chains reduce hallucinations by 60%+

THE EXAMPLE FRAMEWORK - SHOW DON'T TELL
- zero-shot prompts are gambling... few-shot prompts are engineering
- provide 2-3 input/output examples in your desired format
- use this structure: "here's good: [example] here's bad: [counter-example]"
- GPT-5 learns patterns from examples better than from instructions

THE CONSTRAINT SYSTEM - BOUNDARIES CREATE QUALITY
- vague prompts = mediocre outputs (always)
- define what you DON'T want: "avoid generic advice, corporate jargon, obvious statements"
- set output limits: "max 150 words" or "exactly 5 bullet points"
- specify tone explicitly: "write like a mentor, not a textbook"

THE ROLE ENGINEERING - CREATE EXPERT PERSONAS
- "you're an expert" is lazy... be surgical
- use: "you're a senior data scientist who explains complex ML concepts to non-technical founders"
- combine multiple expertise domains for unique angles
- specific roles activate specific training data clusters

THE VALIDATION LAYER - CATCH HALLUCINATIONS
- always request sources or reasoning: "cite your reasoning" or "explain your confidence level"
- ask for alternative perspectives to test accuracy
- add: "what assumptions are you making? what could you be wrong about?"
- GPT-5 can self-critique if you prompt it to

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r/VibeCodersNest • 4d ago
Ok_Gift9191

Vibe Coding: A Beginner's Guide

Tutorials & Guides

Hey there! I put together a quick guide with easy steps to jump into vibe coding.

What is Vibe Coding?

Vibe coding is all about using AI to write code by describing your ideas. Instead of memorizing syntax, you tell the AI what you want (e.g., "Make a webpage with a blue background"), and it generates the code for you. It's like having a junior developer who needs clear instructions but works fast!

Steps to Get Started

1. Pick a tool like Cursor (a VS Code-like editor with AI features) or you might also want to explore Base44, which offers AI-driven coding solutions tailored for rapid prototyping, while Cursor requires installation but has a slick AI chat panel.
2. Start tiny: Begin with something small, like a webpage or a simple script. In Cursor or Base44's editor, create a new file or directory. This gives the AI a canvas to generate code. Base44's platform, for instance, provides pre-built templates to streamline this step.
3. Write a Clear Prompt: The magic of vibe coding happens here. In the AI chat panel (like Base44's code assistant or Cursor's Composer), describe your goal clearly. For example: "Create a webpage that says 'Hello World' with a blue background". Clarity is key.
4. Insert the Code Simply apply the code to your project to see it take shape.
5. Test the Code Run your code to verify it works.
6. Refine and Add Features Rarely is the first output perfect. If it's not quite right, refine your prompt: "Make the text larger and centered." Got an error? Paste it into the AI chat and ask, "How do I fix this?" Tools like Base44's AI assistant are great at debugging and explaining errors. This iterative process is the heart of vibe coding.
7. Repeat the Cycle Build feature by feature, testing each time. You'll learn how the AI translates your words into code and maybe pick up some coding basics along the way.

Example: Building a To-Do List App

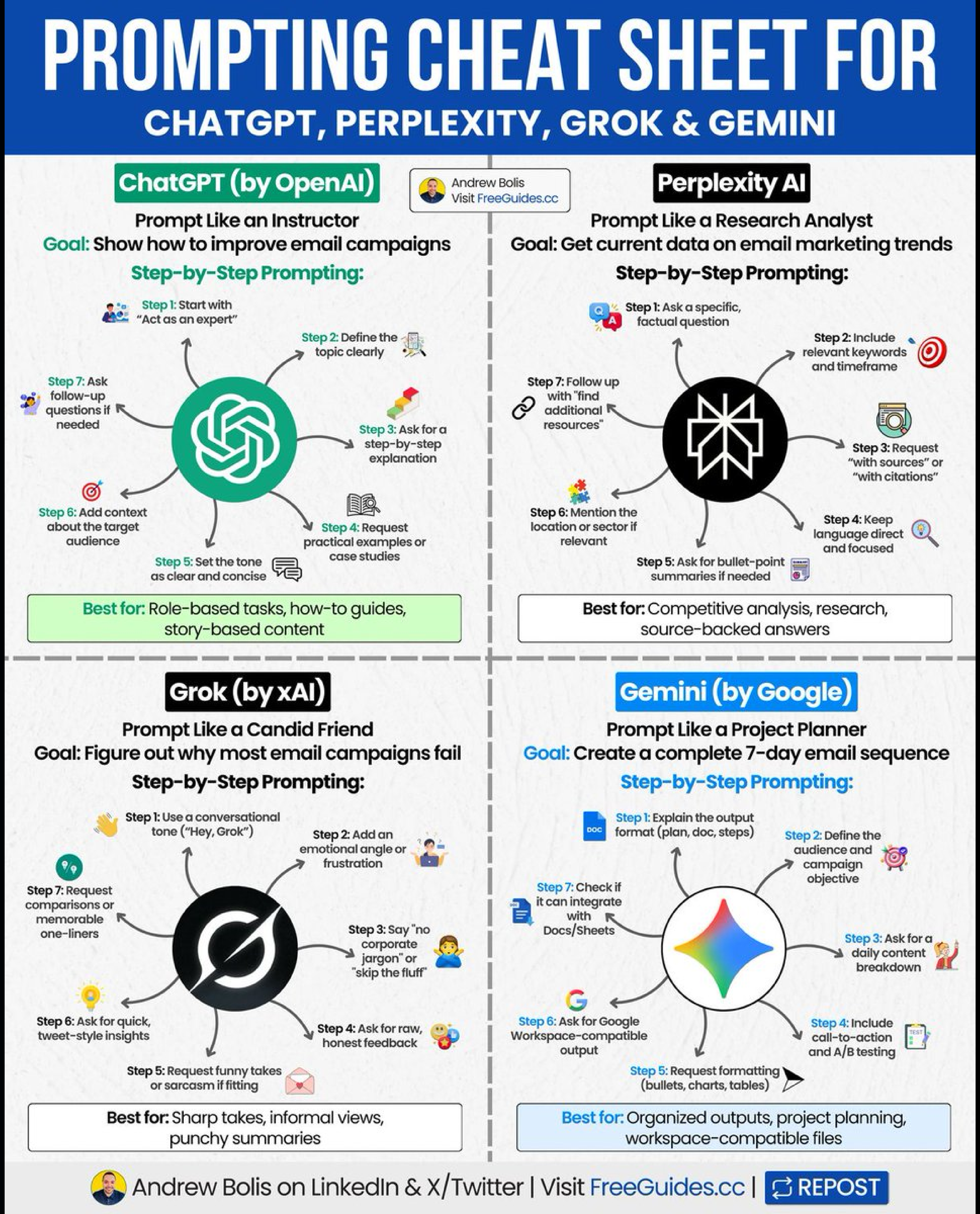
- Prompt 1: "Create an HTML page with an input box, 'Add' button, and task list section" -> AI generates the structure.
- Test: The page loads, but the button is inactive.
- Prompt 2: "When the button is clicked, add the input text to the list and clear the input" -> AI adds JavaScript with an event listener.
- Test: It works, but empty inputs get added.
- Prompt 3: "Don't add empty tasks" -> AI adds a check for empty strings.
- Prompt 4: "Store tasks in local storage to persist after refresh". -> AI implements localStorage. You've now got a working to-do app, all by describing your needs to the AI.

Best Practices for Vibe Coding

- Be Specific: Instead of "Make it pretty", say "Add a green button with rounded corners". Detailed prompts yield better results.
- Start Small: Build a minimal version first, then add features. This works well with platforms like Base44, which support incremental development.
- Review & Test: Always check the AI's code and test frequently to catch bugs early.
- Guide the AI: Treat it like a junior developer- provide clear feedback or examples to steer it.
- Learn as You Go: Ask the AI to explain code to build your understanding.
- Save Your Work: Use versioning to revert if needed.
- Explore Community Resources: Check documentation for templates and tips to enhance your vibe coding experience.

Limitations to Watch For

- Bugs: AI-generated code can have errors or security flaws, so test thoroughly.
- Context: AI may lose track of large projects- remind it of key details or use tools like Base44 that index your code for better context.
- Code Quality: The output might work but be messy- prompt for refactoring if needed.



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r/PromptEngineering • 9 days ago
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After 1000 hours of prompt engineering, I found the 6 patterns that actually matter

Tips and Tricks

I'm a tech lead who's been obsessing over prompt engineering for the past year. After tracking and analyzing over 1000 real work prompts, I discovered that successful prompts follow six consistent patterns.

I call it KERNEL, and it's transformed how our entire team uses AI.

Here's the framework:

K - Keep it simple

- Bad: 500 words of context
- Good: One clear goal
- Example: Instead of "I need help writing something about Redis," use "Write a technical tutorial on Redis caching"
- Result: 70% less token usage, 3x faster responses

E - Easy to verify

- Your prompt needs clear success criteria
- Replace "make it engaging" with "include 3 code examples"
- If you can't verify success, AI can't deliver it
- My testing: 85% success rate with clear criteria vs 41% without

R - Reproducible results

- Avoid temporal references ("current trends", "latest best practices")
- Use specific versions and exact requirements

Executive Summary

- $30–40B spent, 95% zero ROI: Despite massive enterprise investment in GenAI, only 5% of pilots deliver measurable P&L impact.
- The "GenAI Divide": Widespread adoption (tools like ChatGPT, Copilot) vs. low transformation (few integrated, learning systems).
- Barriers: Not regulation, infra, or talent, but learning. Current tools don't retain feedback, adapt, or improve over time.
- Winners: A small set of buyers and vendors succeed by demanding/customizing systems that learn and adapt within workflows.

The Wrong Side of the GenAI Divide

- Adoption ≠ disruption: 7 of 9 sectors show little structural change. Only Tech and Media see meaningful shifts.
- Pilot-to-production gap: 95% of custom enterprise AI tools fail to reach production; only 5% succeed. ChatGPT is widely used but fails for mission-critical workflows.
- Shadow AI economy: Employees use personal ChatGPT/Claude accounts far more than official enterprise tools (90% vs. 40% org purchase rate). This often delivers better ROI.
- Investment bias: ~70% of budgets go to sales/marketing, but biggest ROI lies in back-office automation (procurement, finance, ops).

Crossing the Divide – Builders (Vendors)

- Winning playbook:
  - Focus on narrow, high-value use cases (contract automation, voice AI, code gen).
  - Build systems that learn from feedback and embed deeply into workflows.
  - Secure trust via referrals, partnerships, and integration with existing vendors (channel-driven adoption).
- Timing risk: Window is closing. Enterprises will "lock in" adaptive systems in the next 12–18 months. Switching costs will rise sharply once workflows are trained into tools.

Crossing the Divide – Buyers (Enterprises)

- Build vs. Buy: External partnerships succeed ~2x more than internal builds.
- Best practices:
  - Treat AI vendors like BPOs/consultancies (hold accountable to outcomes, not just demos).
  - Bottom-up adoption: Empower line managers and "prosumers" (frontline users already using ChatGPT) to lead rollouts.
  - Demand deep customization, data security, integration with existing tools, and adaptability.
- ROI lives in overlooked areas:
  - Front-office: Faster lead qualification, better retention.
  - Back-office: $2–10M BPO savings, 30% less agency spend, internalized risk checks.

Workforce Impact

- Not mass layoffs yet: Displacement is targeted (customer support, admin processing, outsourced dev tasks).
- Main impact: Reduced BPO/agency spend and slower hiring growth in disrupted industries (tech, media).
- AI literacy becoming a core hiring filter: Many execs value tool proficiency more than years of experience.

The Next Frontier – Agentic Web

- Beyond agents: The "Agentic Web" will allow autonomous systems to negotiate, transact, and integrate across the internet.
- Shift: From siloed SaaS → interoperable agents using MCP, A2A, and NANDA protocols.
- Outcome: Enterprises move from prompt-driven to protocol-driven automation, where agents self-coordinate across processes.

Core Insights

1. High adoption, low ROI: Most orgs experiment with GenAI, but only a small minority achieve transformation.
2. Learning is the bottleneck: Lack of memory + adaptability prevents workflows from scaling.
3. Shadow AI shows what works: Individual employees are driving ROI faster than official enterprise deployments.
4. Build → Buy shift: External partnerships succeed twice as often; internal builds largely fail.
5. Back office is the hidden goldmine: Real ROI is in BPO elimination and operational cost savings, not front-office marketing hype.
6. The Agentic Web is coming: Enterprises must lock in adaptive, agentic systems now—or risk falling permanently behind.

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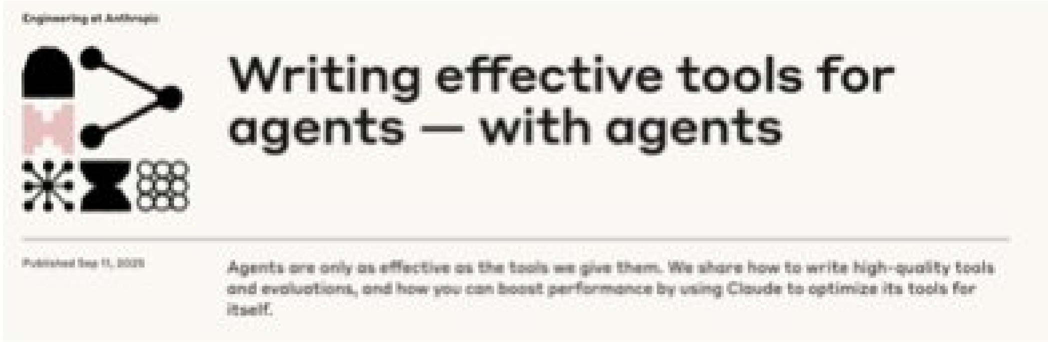

How to Build Better AI Agent Tools (Lessons from Anthropic)

Anthropic recently shared lessons on building tools that agents actually use well. The big idea: agents are not like normal software. They're non-deterministic, have limited context, and rely on tool descriptions to make choices.

If tools are designed with these realities in mind, agents perform better: fewer errors, smarter calls, lower token usage, and more useful outcomes.

What counts as a "tool"

• A tool is a deterministic function an agent can call (e.g. API, search, database query).
• Tools extend what the agent can do, but they must be designed in a way the agent understands.
• The goal is not to expose every endpoint. It's to create the right set of reliable, well-documented tools.

Step 1: Prototype quickly

- Build a minimal version of the tool first.
- Document inputs, outputs, and dependencies clearly.
- Test with a server (like MCP) or extension so agents can call it directly.

Pro tip: let the agent "see" the tool early, even if it's rough. You'll learn faster.

Step 2: Create realistic evaluation tasks

- Use real workflows, not toy problems.
- Add tasks that need multiple tool calls.
- Stress-test edge cases (e.g. missing data, partial failures).

Strong: "Prepare a retention offer for a customer based on logs and profile."
Weak: "Search for purchase_complete in the logs."

Step 3: Run programmatic evaluations

- Automate trials where the agent uses your tools.
- Track not just accuracy, but also:
    • token usage
    • number of tool calls
    • error rates
- Capture agent reasoning to see why it picked certain tools.

Step 4: Iterate with the agent

- Adjust tool names, descriptions, and outputs based on feedback.
- Keep a held-out test set so improvements aren't just overfitting.
- Treat the agent as a collaborator: it can tell you where the tool spec is confusing.

Design principles for effective tools

- Choose the right tools: cover high-impact workflows, not every API.
- Use namespaces: clear naming reduces tool confusion.
- Return context-rich outputs: IDs + labels, not cryptic codes.
- Stay token efficient: truncate, paginate, filter.
- Write precise descriptions: unambiguous input/output names, clear examples.

What to avoid:

- Too many overlapping tools.
- Cryptic or noisy outputs.
- Vague descriptions that mislead the agent.
- Evaluation tasks that are too simple to reveal problems.

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WHICH ONE IS AI ?

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